



ASSIGNMENT 3

VLSI FINAL PROJECT

SURE TRUST

Physical Design Analysis of RISC V processor picorv32 using OpenLane / Sky130

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Assignment Title	Assignment 3 - VLSI Final Project - SURE Trust

Professional formatted submission copy. All original assignment content is retained; front matter, charts, flow diagram, index, and important notes are added for presentation quality.



Professional Index and Analysis Summary

This added front matter improves readability while retaining the original assignment content. It summarizes the design flow, key definitions, important results, and visual interpretations extracted from the existing report data.

Document Index

Section	Purpose	Key Output / Interpretation
Objective and RTL Design	Explains PicoRV32 RTL source setup and directory organization.	RTL files, config.json, base.sdc, pin_order.cfg and OpenLane run folder are defined.
OpenLane Configuration	Lists design name, clock, utilization, aspect ratio, PDN and IO configuration.	60% utilization, 0.7 aspect ratio, 10 ns clock and Metal 4 IO pins are used.
Timing Constraints	Defines SDC constraints for clock, uncertainty and IO delays.	10 ns clock, 1.5 ns uncertainty, 5 ns input and output delay for IO_PCT = 0.5.
Floorplan and Physical-only Cells	Documents floorplan measurements and non-functional physical cells.	146 physical-only cells are reported: 90 tap cells and 56 filler cells.
Placement Analysis	Summarizes buffers, inverters, flip-flops, rows and standard-cell usage.	Placement includes 2,394 buffer/inverter cells and 1,597 flip-flops.
Clock Tree Synthesis	Compares placement and post-CTS clock-buffer structure.	CTS adds high-drive clock buffers for clock distribution.
Routing and Optimization	Compares CTS vs routing cell changes and signal routing observations.	Post-route optimization resizes/replaces several buffer and inverter cells.
IO_PCT Comparison and Power	Compares timing, area, CTS and power between IO_PCT runs.	IO_PCT = 0.5 shows setup violations; IO_PCT = 0.2 timing meets WNS/TNS.

Key Definitions

Term	Definition
RTL	Register Transfer Level description of the digital design using Verilog source files.
OpenLane	Open-source ASIC physical design flow used to run synthesis, floorplanning, placement, CTS, routing and signoff checks.
SKY130	130 nm open-source process design kit / technology node used for standard-cell physical implementation.
SDC	Synopsys Design Constraints file used to define timing requirements such as clock period, input delay, output delay and uncertainty.
Floorplanning	Initial chip/core planning stage where die/core dimensions, utilization, rows, pins and power structure are decided.
Placement	Stage where standard cells are positioned inside the core while optimizing congestion, timing and utilization.
CTS	Clock Tree Synthesis; process of inserting clock buffers/inverters to distribute the clock with controlled skew and latency.
Routing	Physical interconnection stage where nets are connected using metal layers and vias.
WNS / TNS	Worst Negative Slack and Total Negative Slack; timing indicators used to identify setup/hold violations.
PDN	Power Distribution Network; power straps/rings/grid used to deliver VDD/VSS across the chip.



Design Flow Overview

OpenLane / SKY130 Physical Design Flow



The flow converts RTL into a routed layout and verifies design timing, power, and physical implementation quality.

Figure: OpenLane / SKY130 physical implementation flow summarized from the assignment content.

Key Results Snapshot

Metric	Reported Result	Professional Interpretation
Target Clock	10 ns / 100 MHz	Used as the primary timing target for the design.
Clock Uncertainty	1.5 ns	15% timing uncertainty margin is applied.
Core Utilization	60%	Matches the assignment requirement.
Physical-only Cells	146 total	90 tap cells and 56 filler cells support manufacturability and layout continuity.
Placement Buffers + Inverters	2,394 cells	Dominated by clock buffers used for clock distribution.
Flip-Flops	1,597 total	Storage elements are mainly dfxtp_1 cells.
CTS Added Clock Buffers	135 total	119 clkbuf_16 and 16 clkbuf_8 cells were inserted after CTS.
Routing Timing for IO_PCT = 0.5	WNS = -1.28 ns, TNS = -2.48 ns	Setup timing violation is present.
Power Summary	26.9 mW total	Internal power is the largest mW-scale contributor.

Important interpretation: IO_PCT = 0.5 increases input/output delay to 5 ns and leads to setup timing violations in the reported comparison, while IO_PCT = 0.2 remains timing-clean with WNS and TNS reported as 0.00 ns.



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11. Conclusion

The complete physical design flow of the PicoRV32 processor was successfully implemented using the OpenLane framework on the SKY130 technology node. Floorplanning, placement, CTS, routing, and timing verification were completed. The design achieved the specified utilization, implemented the required timing constraints, and generated a routed layout with a propagated clock network. The assignment provided practical experience in ASIC implementation and analysis of physical design stages.

Objective :

1.2 PicoRV32 RTL Design

PicoRV32 is a compact and efficient 32-bit RISC-V processor core developed by Clifford Wolf. It implements the RV32I instruction set architecture and is widely used for research, academic projects, and embedded system applications due to its small area and simple architecture.

The RTL source code of the PicoRV32 processor was obtained from the official GitHub repository maintained by YosysHQ. The complete Verilog source files were downloaded and copied into the OpenLane design directory for physical implementation.



The following command was used to download the RTL source:

```
cd ~/Desktop/Openlane/OpenLane/designs
```

```
git clone https://github.com/YosysHQ/picorv32.git
```

The downloaded Verilog source files were placed inside the **src** directory of the OpenLane design folder so that they could be accessed during synthesis.

1.3 OpenLane Design Directory Structure

The OpenLane flow requires every design to be organized inside a dedicated design directory. The PicoRV32 project directory contains all the files required for synthesis, floorplanning, placement, clock tree synthesis, routing, and timing analysis.

The design directory consists of the following important files.

File	Description
src/	Contains all Verilog RTL source files
config.json	OpenLane configuration file
base.sdc	Timing constraint file
pin_order.cfg	IO pin placement configuration
runs/	Generated during OpenLane execution containing all design results

The complete directory structure is shown below.

```
picorv32/  
├── src/  
├── config.json  
├── base.sdc  
├── pin_order.cfg  
└── runs/
```

OPENLANE DESIGN CONFIGURATION

2.1 Configuration File (config.tcl)

The **config.json** file defines all the design-specific parameters required by OpenLane. It specifies the design name, RTL source files, clock information, floorplanning parameters, power distribution options, and IO pin placement configuration. During the initialization stage, OpenLane reads this file and automatically configures the complete design flow based on the specified parameters.

The configuration used for the PicoRV32 design is shown below.

The description of each configuration parameter is given in Table 2.1.

Parameter	Value	Description
DESIGN_NAME	picorv32	Top-level design name
VERILOG_FILES	dir::src/*.v	RTL source files
CLOCK_PORT	clk	Primary clock input



Parameter	Value	Description
CLOCK_PERIOD	10 ns	Clock period corresponding to 100 MHz
FP_CORE_UTIL	60	Core utilization fixed to 60%
FP_ASPECT_RATIO	0.7	Core aspect ratio
FP_PDN_CORE_RING	true	Enables the power ring around the core
IO_PCT	0.5	Percentage of IO spacing
FP_IO_HMETAL	4	Horizontal IO pins assigned to Metal 4
FP_IO_VMETAL	4	Vertical IO pins assigned to Metal 4
FP_PIN_ORDER_CFG	pin_order.cfg	IO pin placement configuration file

The above configuration satisfies the assignment requirements by setting the core utilization to 60%, aspect ratio to 0.7, enabling the power ring, assigning IO pins to Metal 4, and specifying the custom IO pin placement file.

TIMING CONSTRAINTS

3.1 Timing Constraint File (base.sdc)

The timing constraints for the PicoRV32 processor are defined using the Synopsys Design Constraints (SDC) format. The **base.sdc** file specifies the clock definition, clock uncertainty, input delay, and output delay used during synthesis and timing analysis.

The timing constraint file used in this assignment is shown below.

```
create_clock \  
-name clk \  
-period 10 \  
[get_ports clk]  
  
set_clock_uncertainty 1.5 [get_clocks clk]  
  
set_input_delay 5 \  
-clock clk \  
[all_inputs]  
  
set_output_delay 5 \  
-clock clk \  
[all_outputs]
```

The clock period is specified as 10 ns, corresponding to an operating frequency of 100 MHz. As required by the assignment, the initial clock uncertainty is set to 15% of the clock period (1.5 ns). The input delay and output delay are both fixed to 50% of the clock period (5 ns) to model external timing constraints during synthesis.

The timing constraints applied in the design are summarized in Table 3.1.

Parameter	Value
Clock Period	10 ns
Operating Frequency	100 MHz
Clock Uncertainty	1.5 ns (15%)
Input Delay	5 ns
Output Delay	5 ns



IO PIN CONFIGURATION

4.1 IO Pin Placement Configuration (pin_order.cfg)

The IO pin placement of the PicoRV32 design is controlled using the **pin_order.cfg** file. This file specifies the physical location of each input and output pin on the four sides of the chip.

The four sections used in the configuration file are listed below.

Section	Chip Side
#N	North
#S	South
#E	East
#W	West

North Side (#N)

clk / resetn (System Clock & Active-Low Reset)

mem_ready (Memory acknowledgement)

mem_rdata[31:0] (32-bit Input Data Bus)

pcpi_wr, pcpi_rd[31:0], pcpi_wait, pcpi_ready (Co-processor return interface)

irq[31:0] (Interrupt request lines)

South Side (#S)

trap (CPU Halt/Fault Indicator)

mem_valid / mem_instr (Memory request and Instruction/Data flag)

mem_addr[31:0] (32-bit Memory Address Bus)

mem_wdata[31:0] (32-bit Output Data Bus)

mem_wstrb[3:0] (4-bit Byte Write Strobes)

West Side (#W)



pcpi_valid (Co-processor command valid flag)

pcpi_insn[31:0] (Instruction routed to the co-processor)

pcpi_rs1[31:0] / pcpi_rs2[31:0] (Source Register 1 & 2 values)

eoi[31:0] (End-of-Interrupt outputs)

East Side (#E)

mem_la_read / mem_la_write (Look-Ahead early cycle flags)

mem_la_addr[31:0] (Look-Ahead address bus)

mem_la_wdata[31:0] (Look-Ahead write data bus)

mem_la_wstrb[3:0] (Look-Ahead write byte strobes)

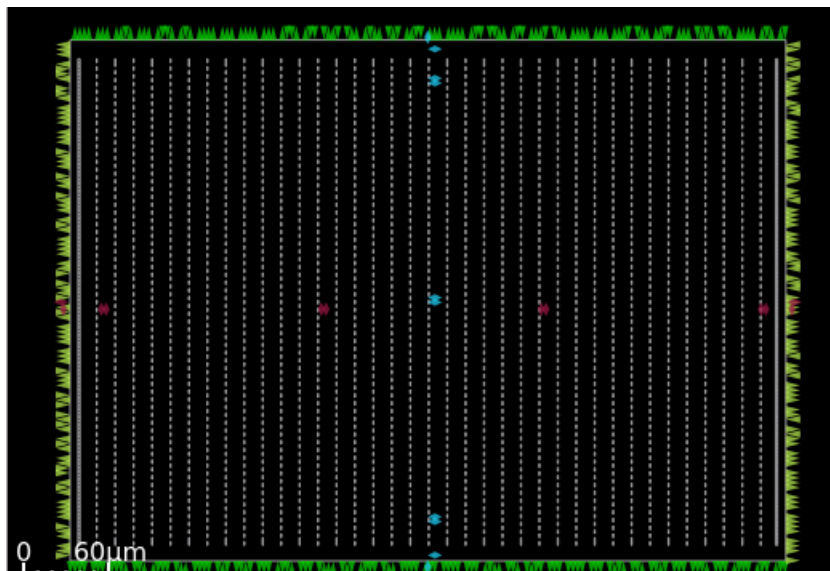
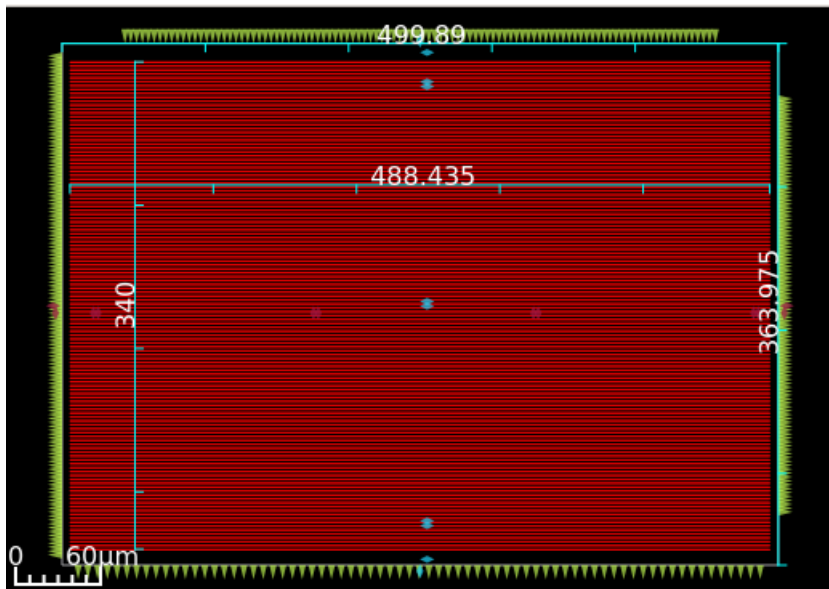
trace_valid / trace_data[35:0] (Hardware tracing/debugging bus)

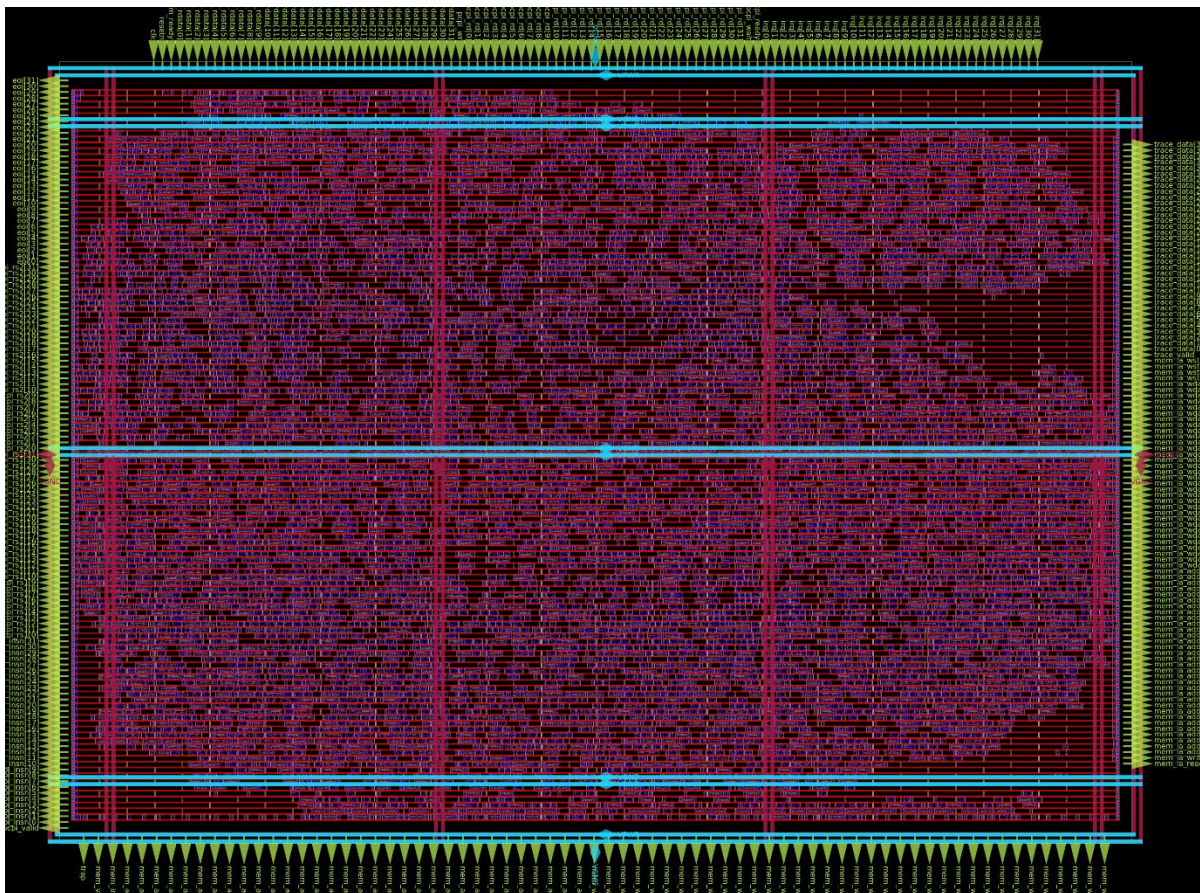
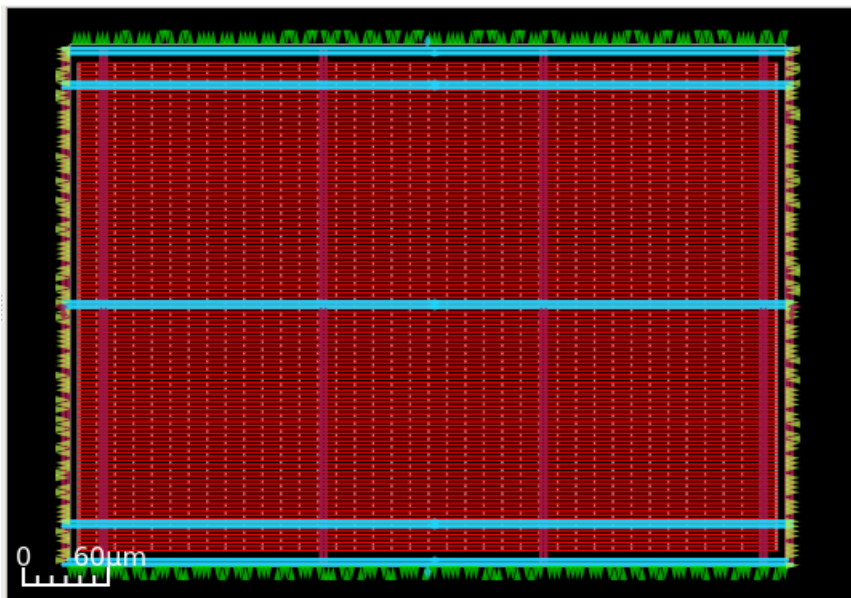
Example :

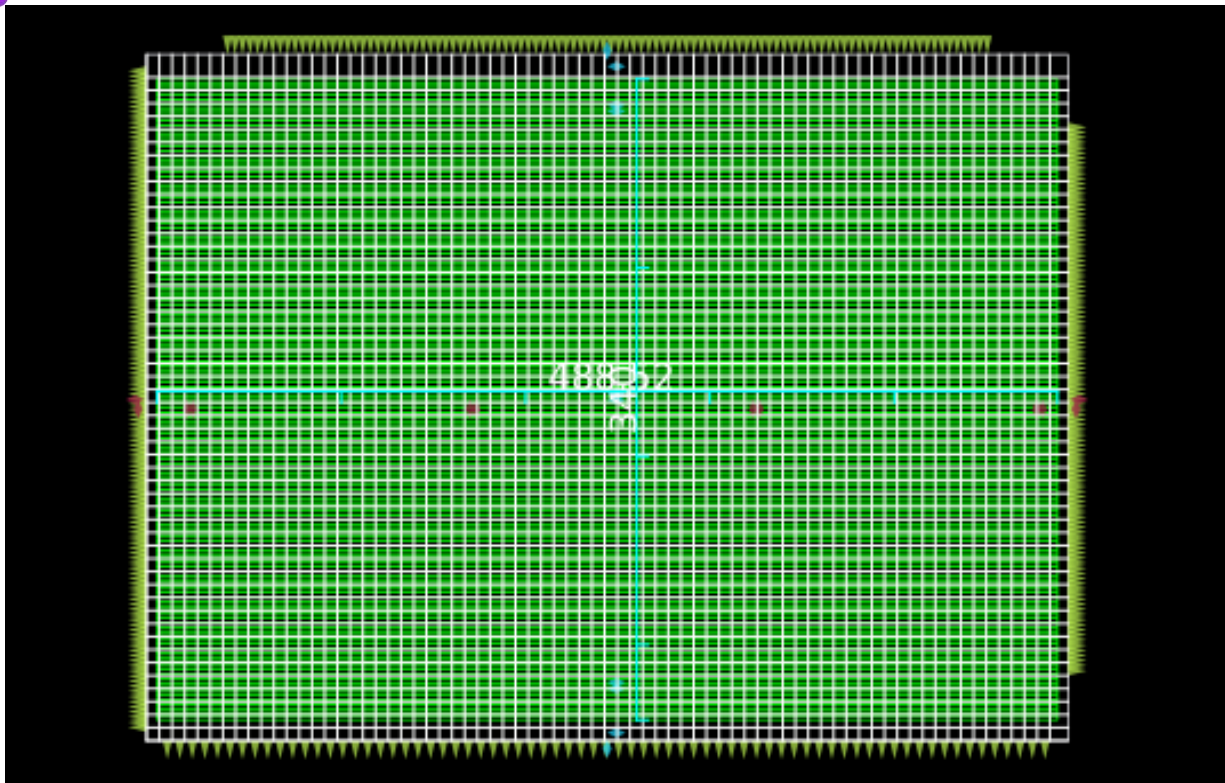
```
#N
^clk$
^resetn$
^mem_ready$
^mem_rdata\[0\]$
^mem_rdata\[1\]$
^mem_rdata\[2\]$
^mem_rdata\[3\]$
^mem_rdata\[4\]$
```

GUI TASK :

Measure the core width and height :

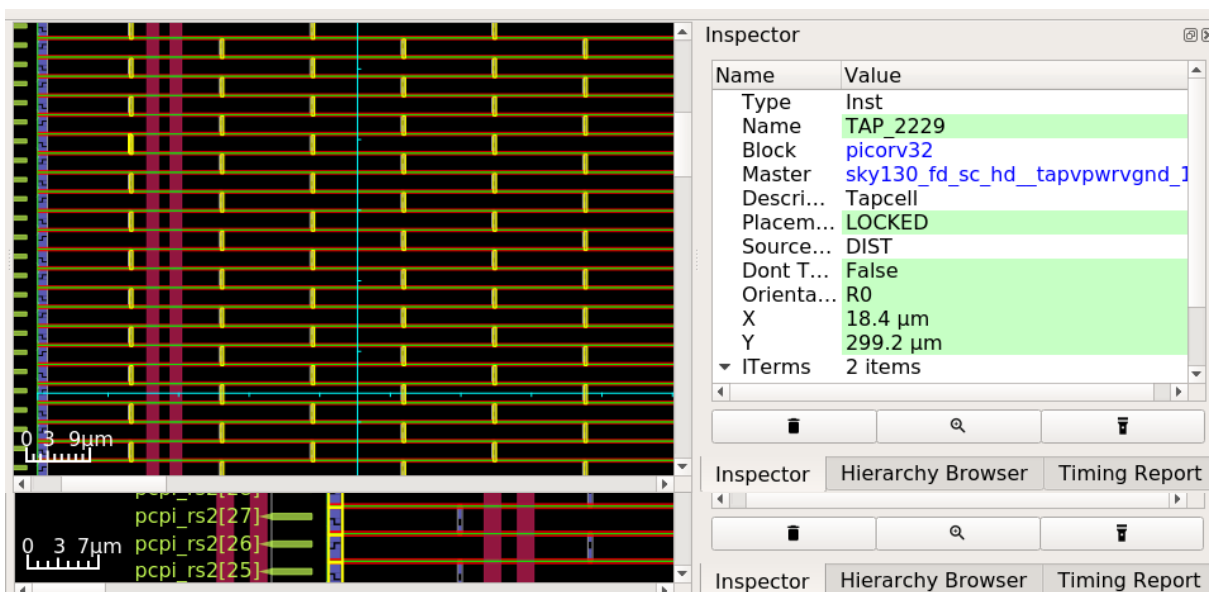






Different types of physical only cells at Floorplan Stage :

▼ ☒ ■ <top>	picorv32 9,789/9,789
▶ Leaf instances	9,789
▼ Physical only	2,599
▶ Fill	250
▶ Tapcell	2,349





PHYSICAL-ONLY CELLS REPORT – 4-BIT COUNTER

Design Name: 4-Bit Counter

Technology: Sky130 HD Standard Cell Library

Stage: Floorplan

Physical-only cells are non-functional cells inserted during the physical design flow. They do not contribute to the logical operation of the circuit but are necessary for manufacturability, power integrity, and layout rule compliance.

Physical-only cells commonly seen in Sky130

- **Tap cells** (sky130_fd_sc_hd__tapvpwrvgnd_*)
 - Connect substrate and well regions to power/ground.
- **Endcap cells** (sky130_fd_sc_hd__decap_*, endcap variants)
 - Placed at row boundaries.
- **Decap cells**
 - Provide local decoupling capacitance.
- **Filler cells** (sky130_fd_sc_hd__fill_*)
 - Fill gaps between standard cells.
- **Tie cells** (sky130_fd_sc_hd__conb_*)
 - Generate constant logic 0 or logic 1.
- **Diode cells** (sky130_fd_sc_hd__diode_*)
 - Used for antenna violation fixes.

1. Tap Cells

- Cell Type: sky130_fd_sc_hd__tapvpwrvgnd
- Count: 90

Purpose:

Tap cells provide connections between the substrate/well regions and the power supply rails (VDD and VSS). They prevent latch-up by maintaining proper biasing of the wells and substrate throughout the chip.

Observation:

A total of 90 tap cells were inserted uniformly across the floorplan to satisfy the maximum tap spacing rules of the Sky130 technology.

2. Filler Cells

- Cell Type: sky130_fd_sc_hd__fill
- Count: 56

Purpose:

Filler cells occupy unused gaps between standard cells after placement. They ensure continuity of power rails and well structures while satisfying fabrication requirements.

Observation:

A total of 56 filler cells were inserted to remove placement gaps and maintain a continuous physical layout.



Summary:

Cell Type	Count
Tap Cells	90
Filler Cells	56

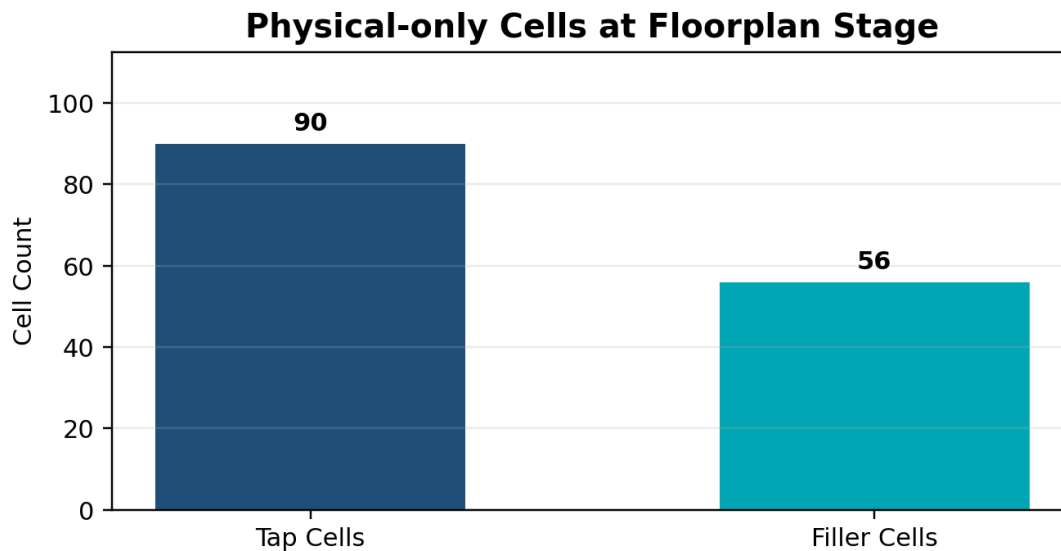


Figure: Tap and filler cell count at the floorplan stage.

Interpretation: Tap cells are the majority of physical-only cells and support well/substrate biasing; filler cells complete layout continuity and power rail connectivity.

Conclusion:

The floorplan of the 4-bit counter contains 146 physical-only cells. Tap cells ensure proper well and substrate connections to prevent latch-up, while filler cells maintain layout continuity and power rail connectivity. Since the design is relatively small, only tap and filler cells were required, and no additional physical-only cells such as diode or decap cells were inserted.

Placement :

Highlight the buffers and inverters in the design :

```
get_cells -filter "ref_name =~ *inv*"
```

```
get_cells -filter "ref_name =~ *buf*"
```

```
>>> get_cells -filter "ref_name =~ *buf*" _705bd13000000000_p_Instance _c05cd13000000000_p_Instance
_305dd13000000000_p_Instance _105ed13000000000_p_Instance _d05fd13000000000_p_Instance
_2068d13000000000_p_Instance _7069d13000000000_p_Instance _206fd13000000000_p_Instance
_5071d13000000000_p_Instance _6074d13000000000_p_Instance _d074d13000000000_p_Instance
_4075d13000000000_p_Instance _e077d13000000000_p_Instance _5078d13000000000_p_Instance
_c078d13000000000_p_Instance _a079d13000000000_p_Instance _f07ad13000000000_p_Instance
_607bd13000000000_p_Instance _d07bd13000000000_p_Instance _407cd13000000000_p_Instance
```



```
_207dd13000000000_p_Instance _007ed13000000000_p_Instance _e07ed13000000000_p_Instance  
_a080d13000000000_p_Instance _f081d13000000000_p_Instance _6082d13000000000_p_Instance  
_d082d13000000000_p_Instance _5086d13000000000_p_Instance _c086d13000000000_p_Instance  
_a087d13000000000_p_Instance _f088d13000000000_p_Instance _6089d13000000000_p_Instance  
_d089d13000000000_p_Instance _408ad13000000000_p_Instance _208bd13000...
```

```
# Find all buffer cells matching the patterns  
set buf_cells [get_cells -hierarchical -filter {ref_name =~ *buf*}]
```

```
# Initialize an array to keep track of counts  
array set buf_counts {}
```

```
# Iterate through each buffer cell, print it, and count it  
foreach b $buf_cells {
```

```
set ref [get_property $b ref_name]  
puts "[get_property $b name] -> $ref"
```

```
if {[info exists buf_counts($ref)]} {  
    incr buf_counts($ref)  
} else {  
    set buf_counts($ref) 1  
}  
}
```

```
# Print the final summary of each buffer type  
puts "\n===== "  
puts "          BUFFER COUNT SUMMARY          "  
puts "===== "  
foreach type [lsort [array names buf_counts]] {  
    puts "$type : $buf_counts($type)"  
}  
puts "===== "
```

and



for inv tcl script :

```
# Find all inverter cells matching the patterns
set inv_cells [get_cells -hierarchical -filter {ref_name =~ *inv*}]
```

```
# Initialize an array to keep track of counts
array set inv_counts {}
```

```
# Iterate through each inverter cell, print it, and count it
foreach i $inv_cells {
```

```
set ref [get_property $i ref_name]
puts "[get_property $i name] -> $ref"
```

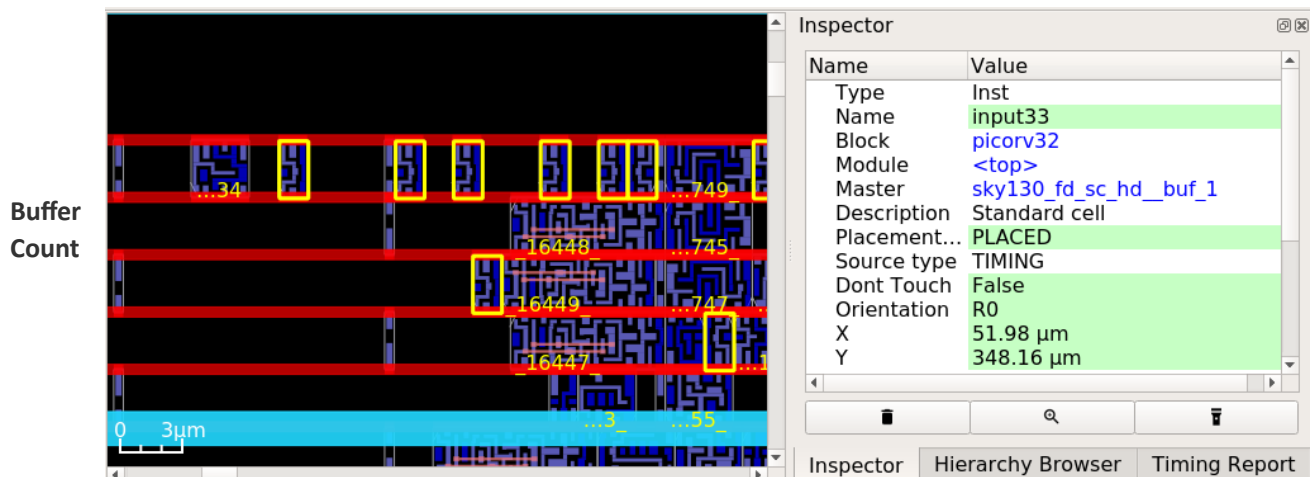
```
if {[info exists inv_counts($ref)]} {
    incr inv_counts($ref)
} else {
    set inv_counts($ref) 1
}
}
```

```
# Print the final summary of each inverter type
puts "\n=====
puts "          INVERTER COUNT SUMMARY          "
puts "=====
foreach type [lsort [array names inv_counts]] {
    puts "$type : $inv_counts($type)"
}
puts "=====
```

The screenshot displays a VLSI design tool interface. On the left, a circuit layout is visible with various components and labels. A yellow box highlights a specific component. On the right, the 'Inspector' window is open, showing the properties of the selected component.

Name	Value
Type	Inst
Name	_09383
Block	picorv32
Module	<top>
Master	sky130_fd_sc_hd_inv_4
Description	Standard cell
Placement...	PLACED
Source type	NONE
Dont Touch	False
Orientation	MX
X	362.48 µm
Y	89.76 µm

Inspector Hierarchy Browser Timing Report



Summary (Placement ODB, IO_PCT = 0.5)

Buffer Type	Count
sky130_fd_sc_hd__buf_1	40
sky130_fd_sc_hd__buf_2	318
sky130_fd_sc_hd__buf_4	231
sky130_fd_sc_hd__buf_6	14
sky130_fd_sc_hd__buf_8	8
sky130_fd_sc_hd__clkbuf_1	1337
sky130_fd_sc_hd__clkbuf_2	26
sky130_fd_sc_hd__clkbuf_4	281
sky130_fd_sc_hd__clkbuf_8	58
sky130_fd_sc_hd__clkbuf_16	0

Total Regular Buffers

- $\text{buf_1} + \text{buf_2} + \text{buf_4} + \text{buf_6} + \text{buf_8}$
- **611**

Total Clock Buffers

- $\text{clkbuf_1} + \text{clkbuf_2} + \text{clkbuf_4} + \text{clkbuf_8}$
- **1702**

Total Buffers

- **2313**

Inverter Count Summary (Placement ODB, IO_PCT = 0.5)

Inverter Type	Count
sky130_fd_sc_hd__clkinv_4	1
sky130_fd_sc_hd__inv_2	79
sky130_fd_sc_hd__inv_4	1



Inverter Type	Count
sky130_fd_sc_hd__inv_6	0
sky130_fd_sc_hd__inv_8	0

Total Inverters

- 81

Overall Summary

Category	Count
Regular Buffers	611
Clock Buffers	1702
Total Buffers	2313
Total Inverters	81
Total Buffer + Inverter Cells	2394

Placement Stage Buffer and Inverter Cell Summary

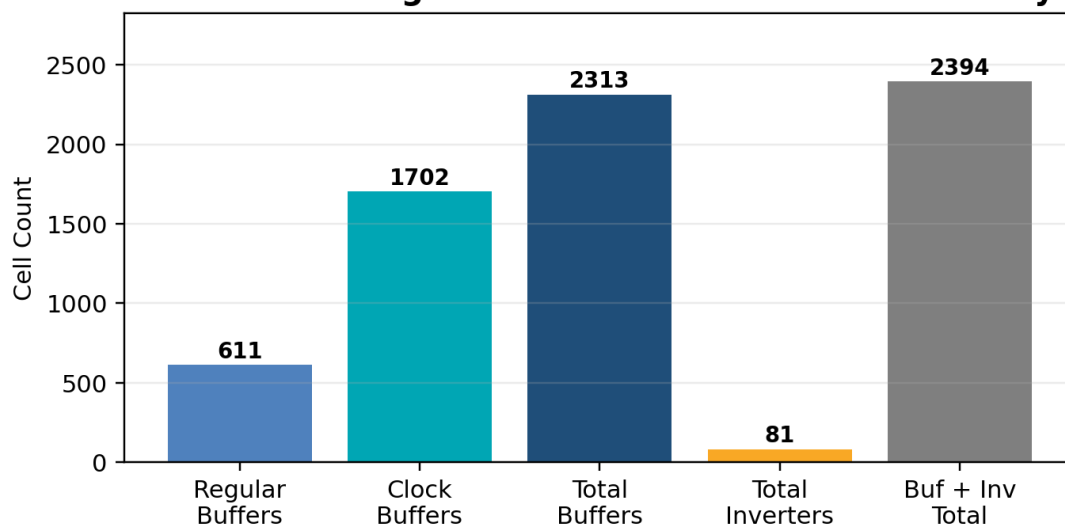


Figure: Placement stage buffer and inverter summary.

Interpretation: Clock buffers dominate the buffer/inverter population, which is expected for a clock-distributed processor implementation with many sequential elements.

Flip-Flop Count Summary

Flip-Flop Type	Drive Strength	Count	Percentage
sky130_fd_sc_hd__dfxtp_1	X1	1430	89.7%
sky130_fd_sc_hd__dfxtp_2	X2	127	8.0%
sky130_fd_sc_hd__dfxtp_4	X4	40	2.5%
Total Flip-Flops	—	1597	100%



Flip-Flop Drive Strength Distribution

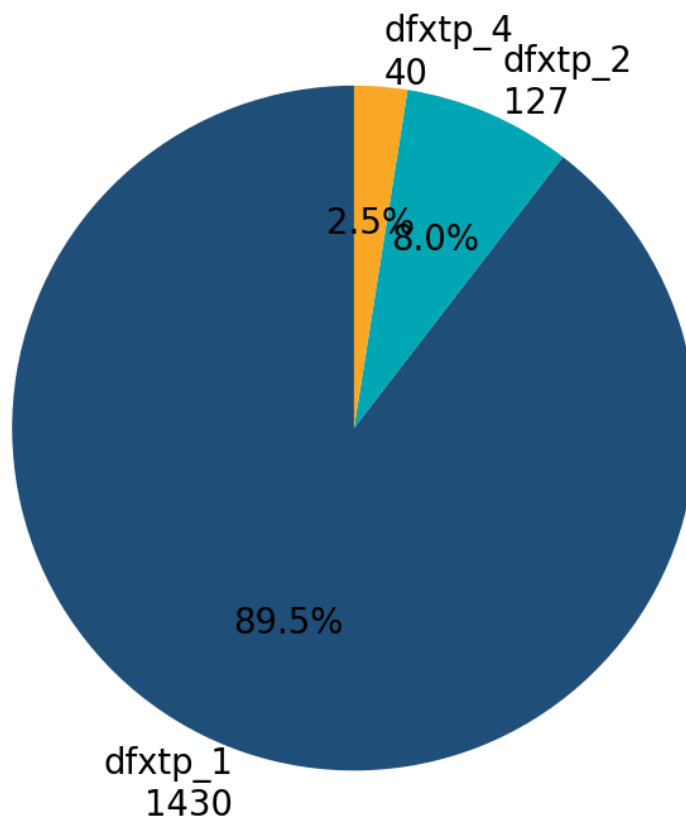


Figure: Flip-flop drive-strength distribution.

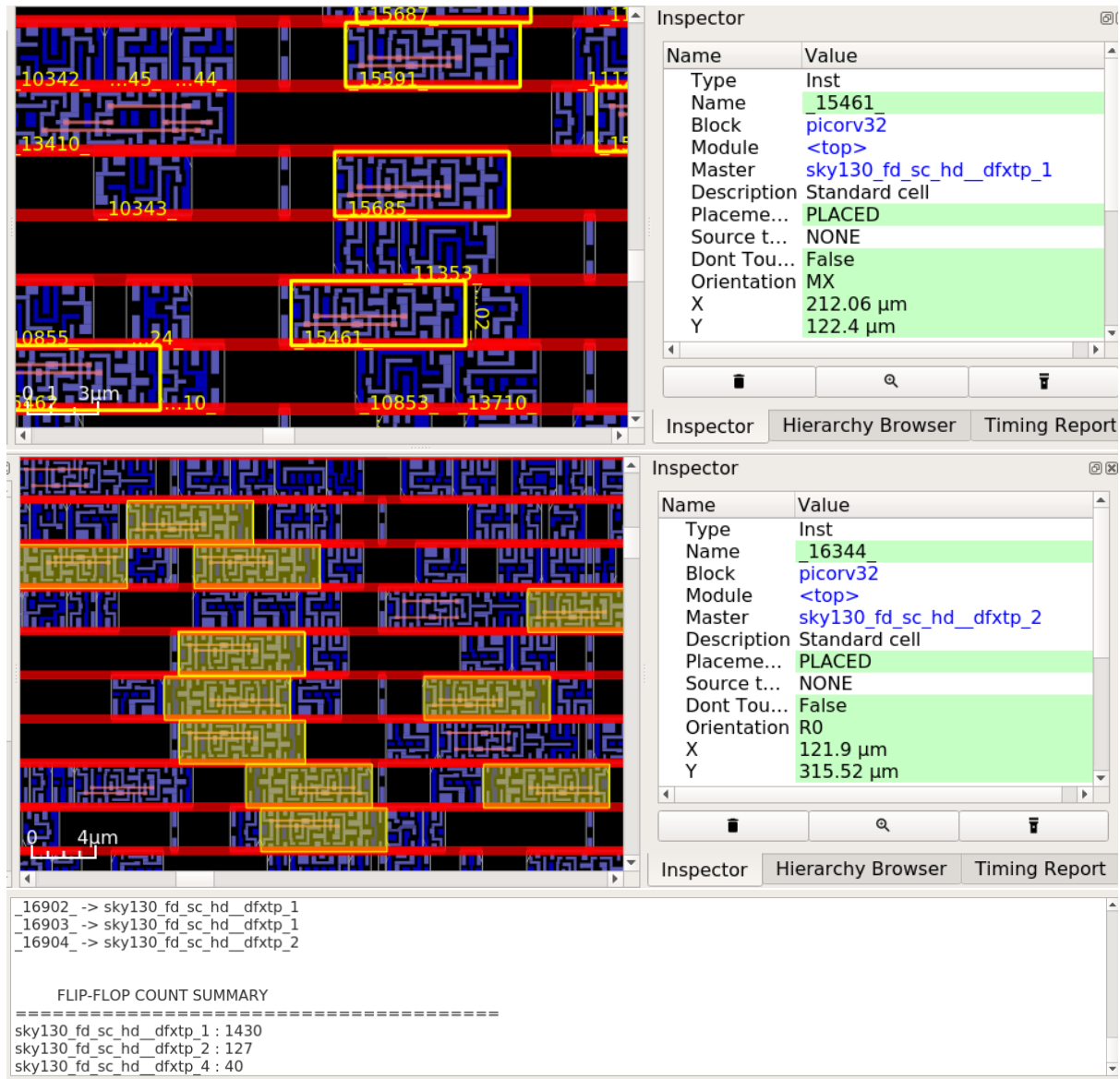
Interpretation: The flip-flop population is strongly dominated by dfxtp_1 cells, indicating that most sequential elements use the smallest drive strength available in the reported placement result.

Observations

1. A total of **1597 D-type positive edge-triggered flip-flops** are present in the placed design before Clock Tree Synthesis (CTS).
2. The design is dominated by **dfxtp_1 flip-flops**, which constitute approximately **89.7%** of all sequential elements. This indicates that most registers drive relatively small loads and do not require higher drive strengths.
3. **dfxtp_2 flip-flops** account for **127 instances (8.0%)** and are used where moderate drive capability is needed to drive larger fanout or longer interconnects.
4. Only **40 instances (2.5%)** of **dfxtp_4** are present, indicating that high-drive flip-flops are required only for a small number of timing-critical or high-fanout paths.
5. The use of predominantly lower drive-strength flip-flops helps reduce **area, dynamic power, and leakage power**, while higher drive-strength variants are selectively used to satisfy timing constraints.

Conclusion

The placed design contains **1597 flip-flops**, with **dfxtp_1** being the predominant type (**89.7%**). This distribution indicates an optimized design that primarily uses low-drive sequential cells for power and area efficiency, while reserving higher drive-strength flip-flops (dfxtp_2 and dfxtp_4) for critical timing paths and higher fanout requirements. This is a typical characteristic of a timing-aware synthesized and placement-optimized digital design before CTS.



ROW ROW_0 unithd 5520 10880 N DO 1062 BY 1 STEP 460 0 ;
ROW ROW_1 unithd 5520 13600 FS DO 1062 BY 1 STEP 460 0 ;
ROW ROW_2 unithd 5520 16320 N DO 1062 BY 1 STEP 460 0 ;
...

Site Row Data

Parameter	Value	Derivation
Site Type	unithd	Standard-cell site used by SKY130 HD library
Starting X-coordinate	5520 DBU	From the DEF file
Starting Y-coordinate (Row 0)	10880 DBU	From the DEF file
Site Width	460 DBU (0.46 µm)	STEP 460 0
Site Height (Row Height)	2720 DBU (2.72 µm)	13600 – 10880 = 2720 DBU
Sites per Row	1062	DO 1062



Parameter	Value	Derivation
Row Width	488520 DBU (488.52 μm)	1062 $\times$ 460 DBU
Row Orientation	Alternating N and FS	As shown in the DEF

Row Orientation

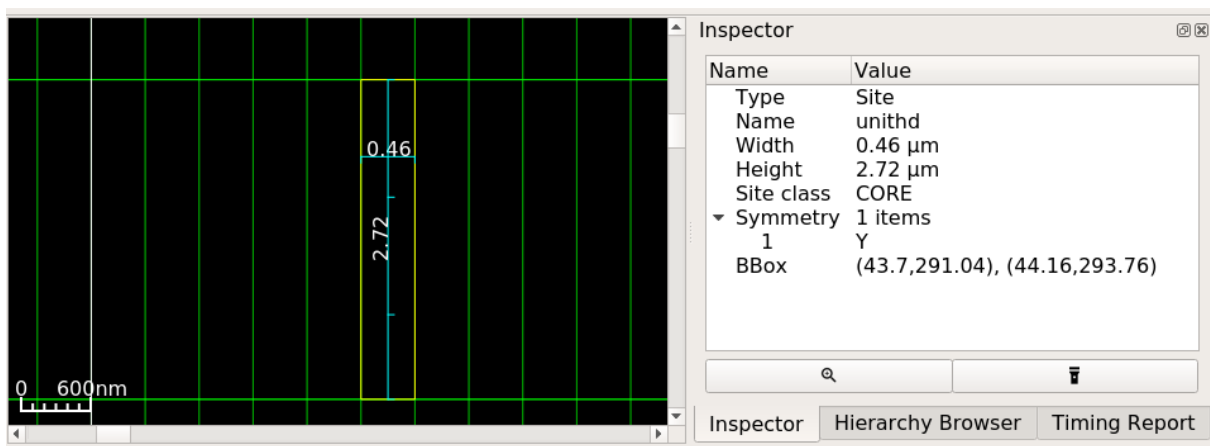
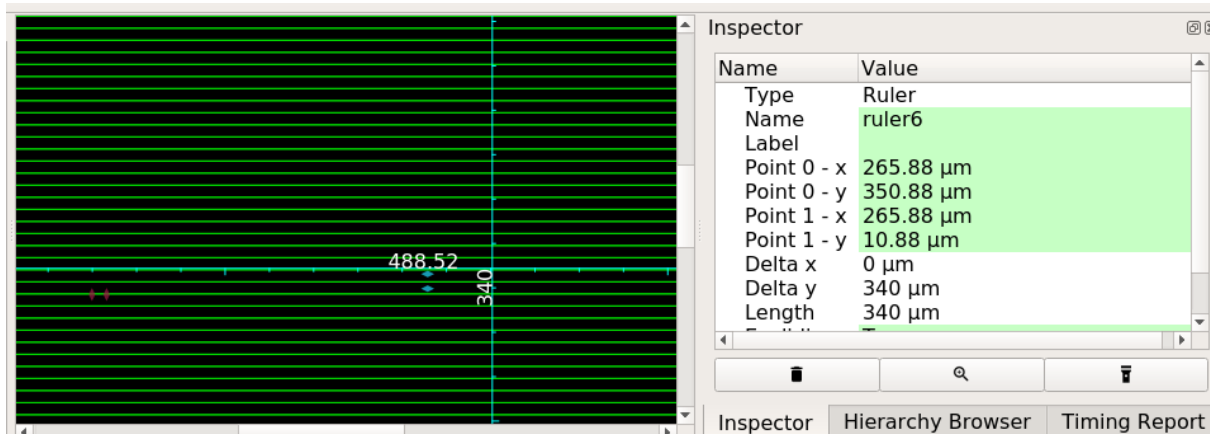
Row	Orientation
ROW_0	N
ROW_1	FS
ROW_2	N
ROW_3	FS
ROW_4	N
ROW_5	FS
...	Alternating

Observation

The placement rows are created using the **unithd** standard-cell site of the SKY130 HD library. Each site has a **fixed width of 0.46 μm** and a **fixed height of 2.72 μm** . Every row contains **1062 placement sites**, resulting in a total row width of **488.52 μm** . The rows are arranged in an alternating **N (Normal)** and **FS (Flipped South)** orientation. This alternating arrangement ensures proper alignment of the power rails (VDD and VSS) between adjacent rows, allowing standard cells to be placed legally while maintaining correct power connectivity. The uniform row dimensions provide a regular placement grid that simplifies cell placement, routing, and timing optimization during the physical design flow.

Key Values for Your Report

- **Site Type:** unithd
- **Site Width:** 0.46 μm
- **Site Height:** 2.72 μm
- **Sites per Row:** 1062
- **Row Width:** 488.52 μm
- **Orientation:** Alternating N / FS
- **Purpose:** Supports legal standard-cell placement with proper VDD/VSS rail alignment and efficient routing.



Different Library cells :

Different Library Cells Used in the Design

Library Cell Category	Cell Types Used	Count
Buffers	buf_1, buf_2, buf_4, buf_6, buf_8	611
Clock Buffers	clkbuf_1, clkbuf_2, clkbuf_4, clkbuf_8	1702
Clock Inverters	clkinv_4	1
Inverters	inv_2, inv_4	80
Flip-Flops	dfxtp_1, dfxtp_2, dfxtp_4	1597
Multiplexers	mux2_*, mux4_*	2152
AND Gates	and2_*, and3_*, and4_*, and2b_*, and3b_*, and4b_*, and4bb_*	522
OR Gates	or2_*, or3_*, or4_*, or2b_*, or3b_*, or4b_*	481
NAND Gates	nand2_*, nand3_*, nand4_*, nand2b_*, nand3b_*	258



Library Cell Category	Cell Types Used	Count
NOR Gates	nor2_*, nor3_*, nor4_*, nor3b_*, nor4b_*	233
XOR/XNOR Gates	xor2_*, xnor2_*	119
AOI/OAI & Complex Gates	a21*, a22*, a31*, a32*, a211*, a221*, a311*, o21*, o22*, o31*, o211*, o221*, etc.	~1480
Constant Cells	conb_1	106
Decoupling Capacitors	decap_3	250
Delay Cells	dlymetal6s2s_1	17
Tap Cells	tapvpwrvrgnd_1	2349

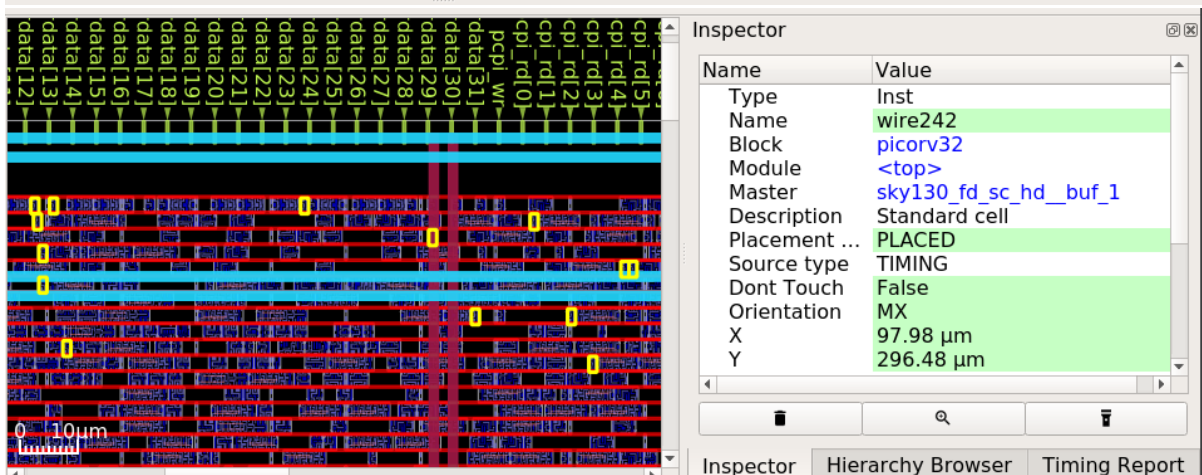
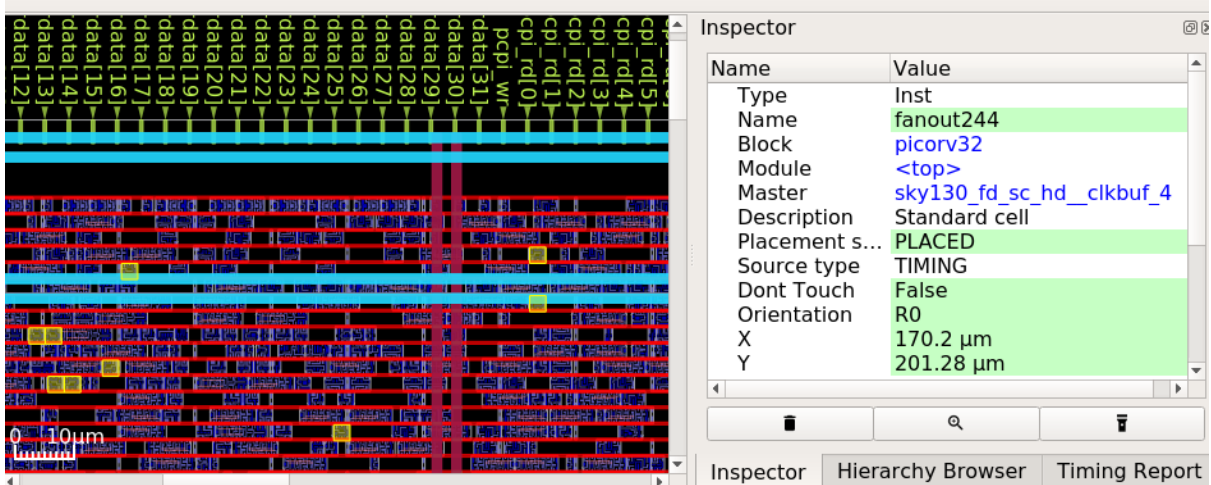
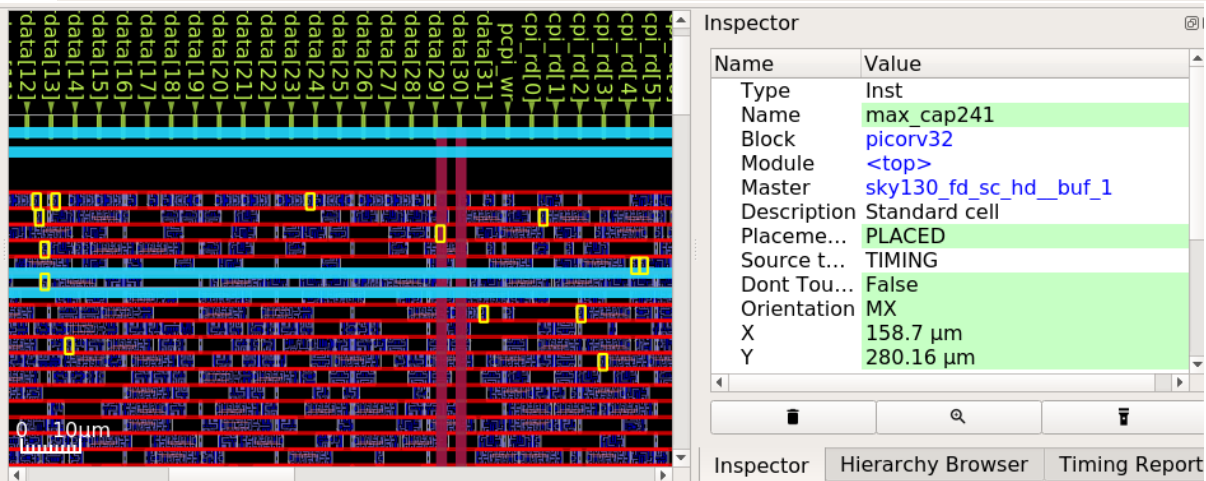
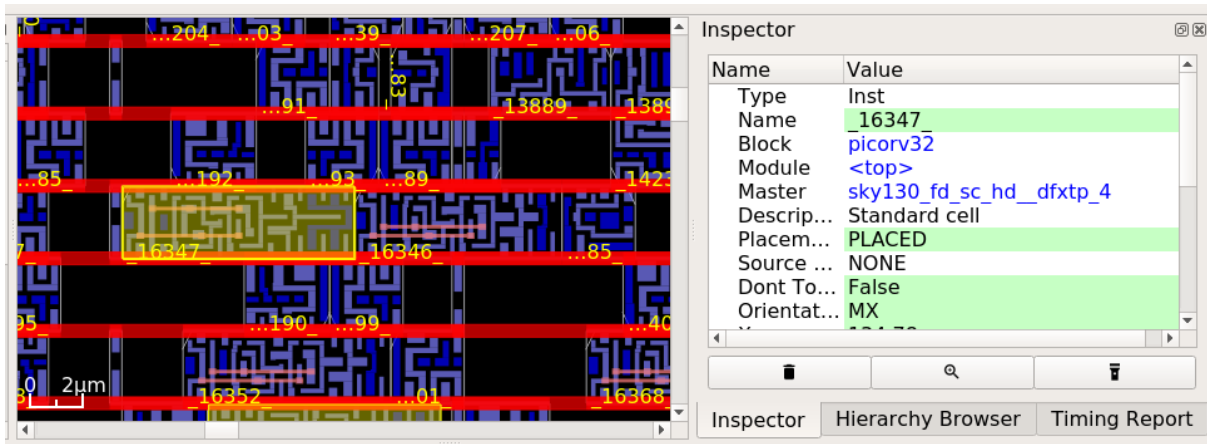
Total Standard Cells = 12,254

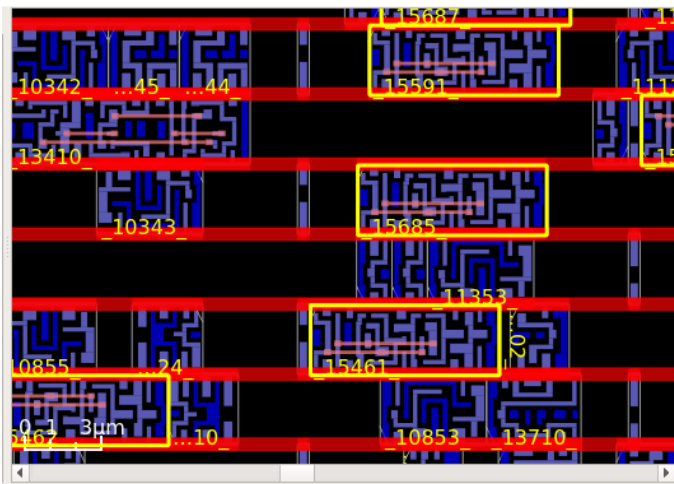
Observations

1. The design consists of **12,254 standard cells** from the **SKY130 HD standard cell library**.
2. **Tap cells (2,349 instances)** form the largest physical-only cell category, ensuring proper substrate and well connections throughout the layout.
3. **Multiplexers (2,152 instances)** and **flip-flops (1,597 instances)** occupy a significant portion of the design, indicating a highly sequential architecture with extensive data selection logic.
4. The design contains **1,702 clock buffers**, reflecting the complexity of the clock distribution network.
5. Most logic gates use the **drive strength _1** variant, while higher drive strengths (**_2, _4, _6, _8**) are used selectively in timing-critical paths.
6. The presence of **buffers, inverters, decoupling capacitors, delay cells, and tap cells** demonstrates that the design has been optimized for timing, signal integrity, and manufacturability.

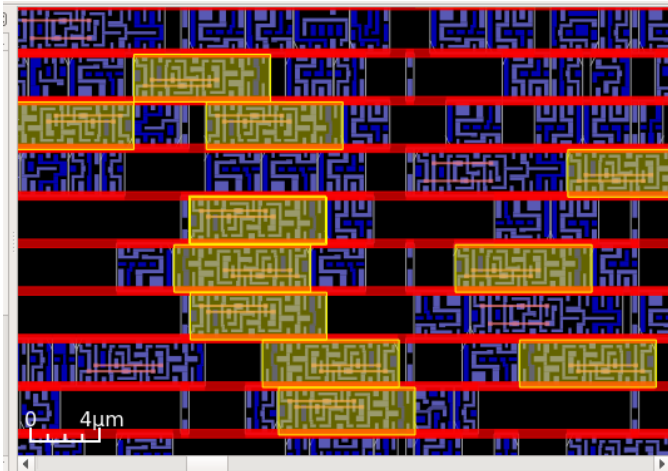
Conclusion

The implemented design uses a diverse set of standard cells from the **SKY130 HD library**, including combinational logic gates, sequential elements, multiplexers, buffers, inverters, clock buffers, and physical-only cells such as tap cells and decoupling capacitors. The majority of logic cells employ lower drive strengths to minimize area and power consumption, while higher drive-strength cells are selectively used to satisfy timing and fanout requirements. The substantial number of clock buffers and tap cells indicates a well-optimized physical implementation that balances timing performance, power distribution, and layout reliability.

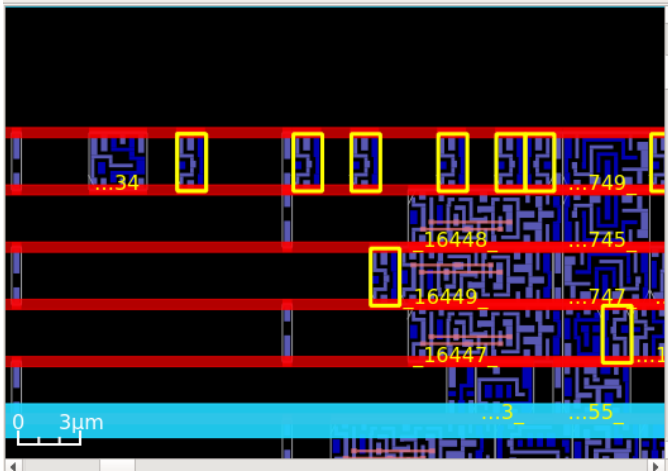




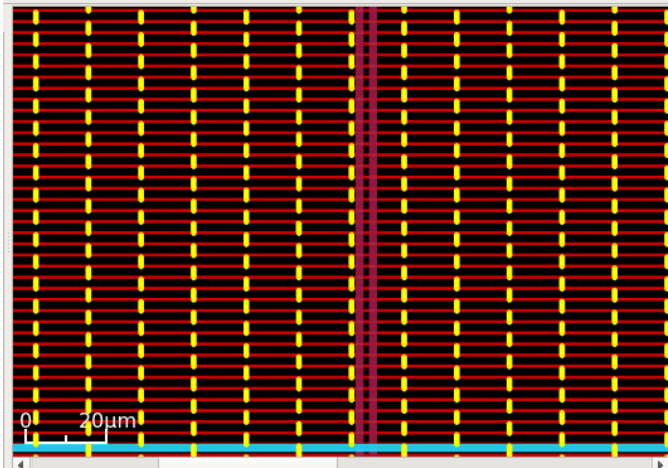
Name	Value
Type	Inst
Name	15461
Block	picorv32
Module	<top>
Master	sky130_fd_sc_hd_dfxt1_1
Description	Standard cell
Placeme...	PLACED
Source t...	NONE
Dont Tou...	False
Orientation	MX
X	212.06 μm
Y	122.4 μm



Name	Value
Type	Inst
Name	16344
Block	picorv32
Module	<top>
Master	sky130_fd_sc_hd_dfxt2_2
Description	Standard cell
Placeme...	PLACED
Source t...	NONE
Dont Tou...	False
Orientation	R0
X	121.9 μm
Y	315.52 μm



Name	Value
Type	Inst
Name	input33
Block	picorv32
Module	<top>
Master	sky130_fd_sc_hd_buf1_1
Description	Standard cell
Placement...	PLACED
Source type	TIMING
Dont Touch	False
Orientation	R0
X	51.98 μm
Y	348.16 μm



Name	Value
Type	Inst
Name	TAP_1809
Block	picorv32
Master	sky130_fd_sc_hd_tapvpwrvrgnd_1
Descri...	Tapcell
Placem...	LOCKED
Source...	DIST
Dont T...	False
Orienta...	MX
X	160.08 μm
Y	236.64 μm
ITerms	2 items



Inspector

Name	Value
Type	Inst
Name	12123
Block	picorv32
Module	<top>
Master	sky130_fd_sc_hd_and2_1
Descrip...	Standard cell
Placem...	PLACED
Source ...	NONE
Dont To...	False
Orientat...	R0
X	261.28 μm
Y	315.52 μm

Inspector Hierarchy Browser Timing Report

Inspector

Name	Value
Type	Inst
Name	12135
Block	picorv32
Module	<top>
Master	sky130_fd_sc_hd_nor2_1
Descrip...	Standard cell
Placem...	PLACED
Source ...	NONE
Dont To...	False
Orientat...	MY
X	264.04 μm
Y	315.52 μm

Inspector Hierarchy Browser Timing Report

Inspector

Name	Value
Type	Inst
Name	12145
Block	picorv32
Module	<top>
Master	sky130_fd_sc_hd_xor2_1
Description	Standard cell
Placeme...	PLACED
Source t...	NONE
Dont Tou...	False
Orientation	R0
X	266.8 μm
Y	315.52 μm

Inspector Hierarchy Browser Timing Report

Inspector

Name	Value
Type	Inst
Name	12156
Block	picorv32
Module	<top>
Master	sky130_fd_sc_hd_inv_2
Descrip...	Standard cell
Placem...	PLACED
Source ...	NONE
Dont To...	False
Orientat...	MX
X	270.94 μm
Y	318.24 μm

Inspector Hierarchy Browser Timing Report



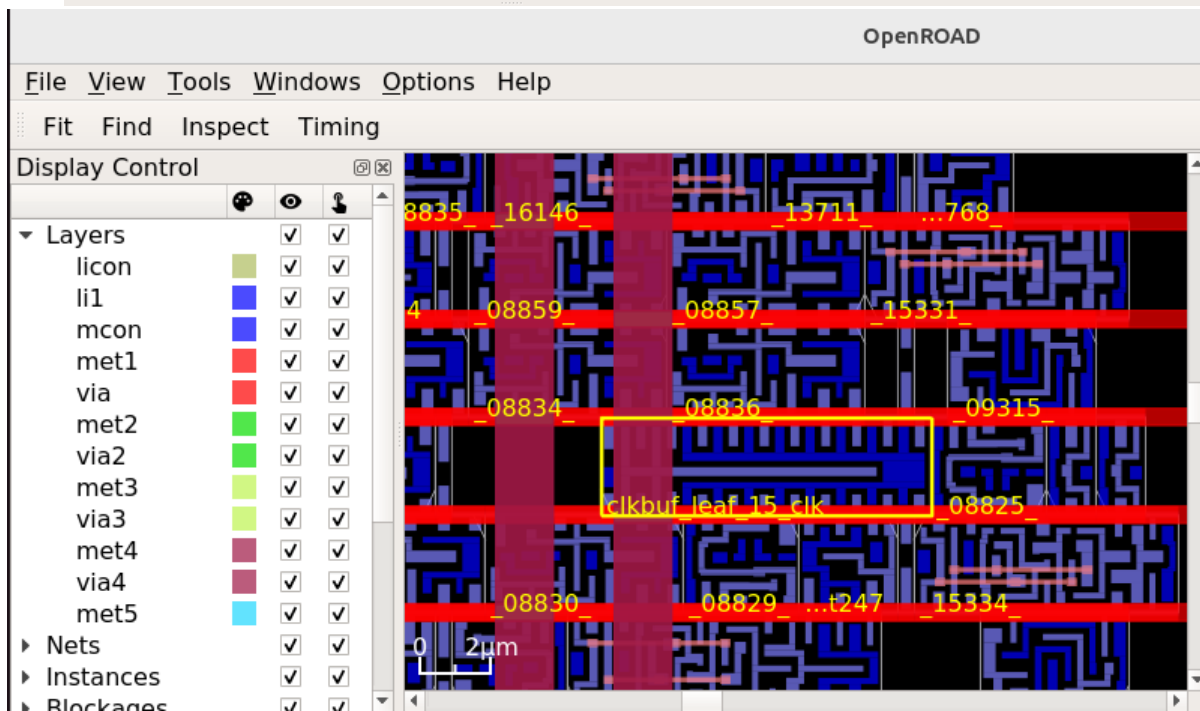
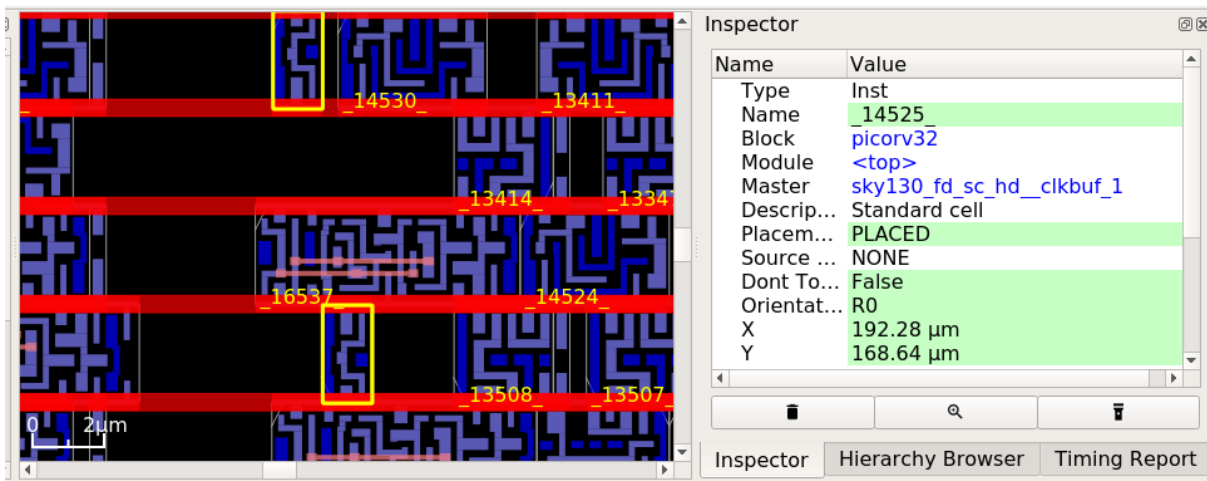
The figure displays four screenshots of a VLSI design tool, each showing a different instance of a 'picorv32' block. The blocks are highlighted with yellow boxes, and their properties are listed in the 'Inspector' panel on the right of each screenshot.

Screenshot 1: The block is named '09383'. Its properties are: Type: Inst, Name: 09383, Block: picorv32, Module: <top>, Master: sky130_fd_sc_hd_inv_4, Description: Standard cell, Placement: PLACED, Source type: NONE, Dont Touch: False, Orientation: MX, X: 362.48 μm , Y: 89.76 μm .

Screenshot 2: The block is named '09434'. Its properties are: Type: Inst, Name: 09434, Block: picorv32, Module: <top>, Master: sky130_fd_sc_hd_nor4_1, Description: Standard cell, Placement st...: PLACED, Source type: NONE, Dont Touch: False, Orientation: MY, X: 80.5 μm , Y: 315.52 μm .

Screenshot 3: The block is named '12759'. Its properties are: Type: Inst, Name: 12759, Block: picorv32, Module: <top>, Master: sky130_fd_sc_hd_o21a_1, Description: Standard cell, Placement...: PLACED, Source type: NONE, Dont Touch: False, Orientation: MX, X: 439.3 μm , Y: 149.6 μm .

Screenshot 4: The block is named '13678'. Its properties are: Type: Inst, Name: 13678, Block: picorv32, Module: <top>, Master: sky130_fd_sc_hd_a211o_1, Description: Standard cell, Placeme...: PLACED, Source t...: NONE, Dont Tou...: False, Orientation: MX, X: 160.54 μm , Y: 122.4 μm .



Instance	Cell Type
_5000ca3e00000000_p_Instance	sky130_fd_sc_hd__clkbuf_2
_c000ca3e00000000_p_Instance	sky130_fd_sc_hd__buf_2
_3001ca3e00000000_p_Instance	sky130_fd_sc_hd__clkbuf_4
_a001ca3e00000000_p_Instance	sky130_fd_sc_hd__clkbuf_4
_1002ca3e00000000_p_Instance	sky130_fd_sc_hd__clkbuf_4

```
>>> get_cells -filter "ref_name =~ *buf*"
... get_cells -filter "ref_name =~ *inv*"
_a06bba3000000000_p_Instance
>>> get_property _a06bba3000000000_p_Instance ref_name
sky130_fd_sc_hd__inv_2
```



The screenshot shows the OpenROAD GUI with a routing layout. The central canvas displays a complex routing pattern with various layers and components. The left sidebar shows the 'Display Control' panel with a list of layers (licon, li1, mcon, met1, via, met2, via2, met3, via3, met4, via4, met5) and their visibility status. The right sidebar shows the 'Inspector' panel with a table of properties for the selected instance.

Name	Value
Type	Inst
Name	11
Block	up_counter
Module	<top>
Master	sky130_fd_sc_hd_inv_2
Description	Standard cell
Placement sta...	PLACED
Source type	NONE
Dont Touch	False
Orientation	R180
X	57.5 µm
Y	46.24 µm

The bottom status bar shows the coordinates 60.841, 43.852.

The screenshot shows the OpenROAD GUI with a routing layout. The central canvas displays a complex routing pattern with various layers and components. The left sidebar shows the 'Display Control' panel with a list of layers (licon, li1, mcon, met1, via, met2, via2, met3, via3, met4, via4, met5) and their visibility status. The right sidebar shows the 'Inspector' panel with a table of properties for the selected instance.

Name	Value
Type	Inst
Name	output3
Block	up_counter
Module	<top>
Master	sky130_fd_sc_hd_clkbuf_4
Description	Standard cell
Placement sta...	PLACED
Source type	TIMING
Dont Touch	False
Orientation	MY

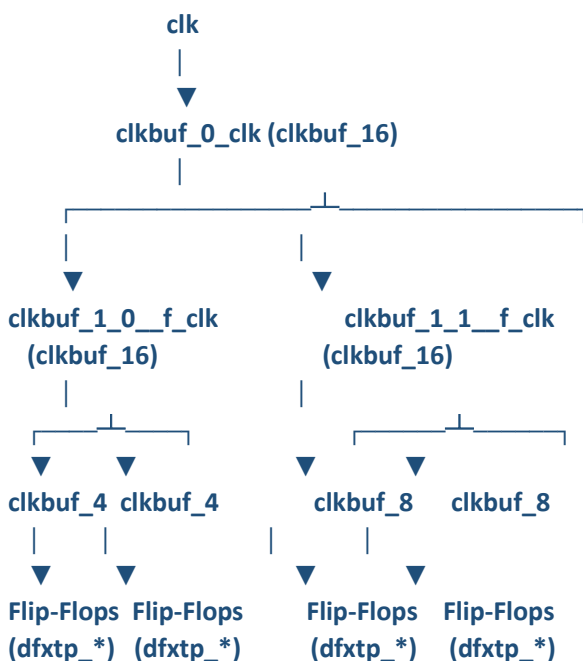
The bottom status bar shows the coordinates 8.774, 10.825.



Height and width of different library cells :

CTS:

Highlight Representative Clock Tree Structure (PicoRV32)



Root Clock Buffer

Instance	Cell Type	Coordinates
clkbuf_0_clk	clkbuf_16	(47380, 54400)

Observation

- The root clock buffer is a high-drive clkbuf_16 placed near the center of the design.
- It serves as the main driver of the clock tree and distributes the clock signal to multiple branches.
- Using a high-drive buffer at the root minimizes delay while driving a large fanout.

Branch Clock Buffers

Instance	Cell Type	Coordinates
clkbuf_1_0__f_clk	clkbuf_16	(39100, 24480)
clkbuf_1_1__f_clk	clkbuf_16	(36340, 51680)

Observation

- The branch buffers divide the clock into separate regions of the chip.



- Each branch drives a subset of flip-flops, thereby reducing the capacitive load seen by the root buffer.
- This hierarchical distribution minimizes clock skew and improves timing reliability.

CTS Buffer Insertion Analysis

Buffer Count Comparison

Cell Type	Placement	After CTS	Change
clkbuf_1	1337	1337	0
clkbuf_2	26	26	0
clkbuf_4	281	281	0
clkbuf_8	58	74	+16
clkbuf_16	0	119	+119

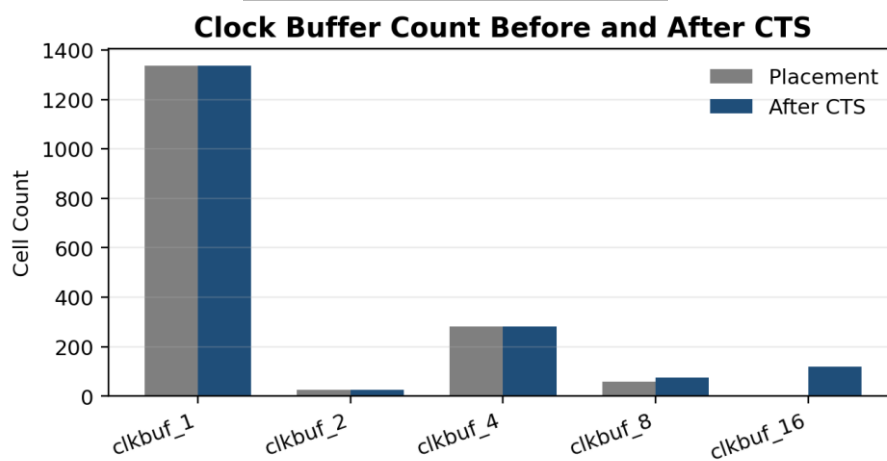


Figure: Clock buffer count before and after CTS.

Interpretation: CTS mainly adds high-drive clock buffers, especially clkbuf_16, to build a balanced and stronger clock distribution network.



Observation

- CTS inserted 119 new clkbuf_16 cells and 16 additional clkbuf_8 cells.
- No existing clkbuf_1, clkbuf_2, or clkbuf_4 cells were modified.
- The inserted high-drive buffers form the backbone of the clock distribution network and are strategically placed to drive long interconnects and high-fanout branches.

CTS Inverter Analysis

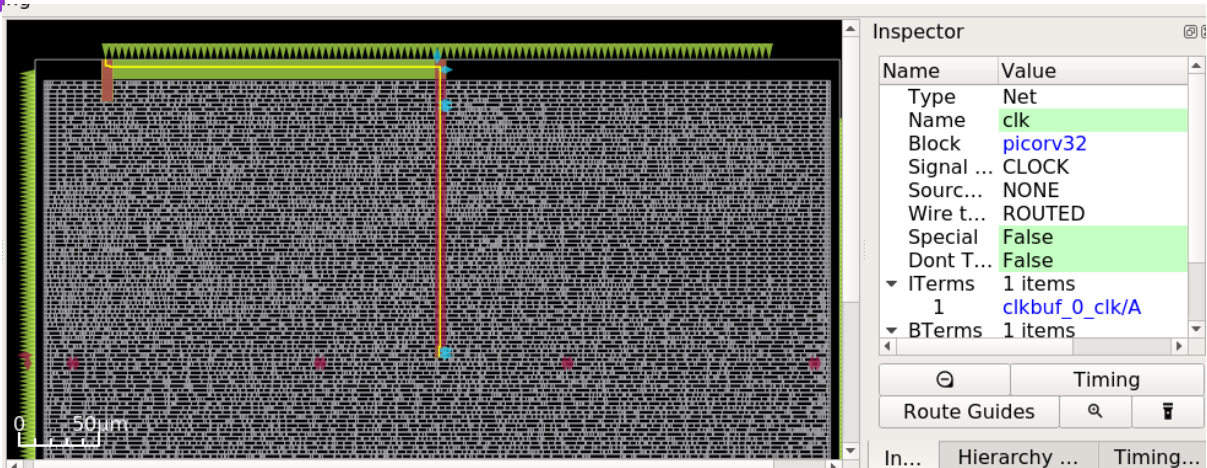
Cell Type	Placement	CTS	Change
clkinv_4	1	1	0
inv_2	79	79	0
inv_4	1	1	0

Observation

- The number of inverter cells remained unchanged after CTS.
- No additional clock inverters were required because the clock polarity was already correct.
- CTS achieved the required clock balancing solely through buffer insertion without modifying the clock polarity.

Overall CTS Observation

Clock Tree Synthesis inserted 135 additional clock buffers (119 clkbuf_16 and 16 clkbuf_8) while leaving the existing lower drive-strength clock buffers and all inverter cells unchanged. The newly inserted clkbuf_16 cells serve as the root and branch buffers of the clock tree, creating a hierarchical distribution network that efficiently drives the clock to all sequential elements. This strategy minimizes clock skew, reduces insertion delay, balances fanout across the design, and improves overall timing performance without requiring additional clock inversions.



Routing :

Newly added Buffer and inv :

Buffer Comparison (CTS → Routing)

Buffer Type	After CTS	After Routing	Change
buf_1	40	40	0
buf_2	318	109	-209
buf_4	231	284	+53
buf_6	14	37	+23
buf_8	8	17	+9
clkbuf_1	1337	1333	-4
clkbuf_2	26	26	0
clkbuf_4	281	387	+106
clkbuf_8	74	102	+28
clkbuf_16	119	121	+2

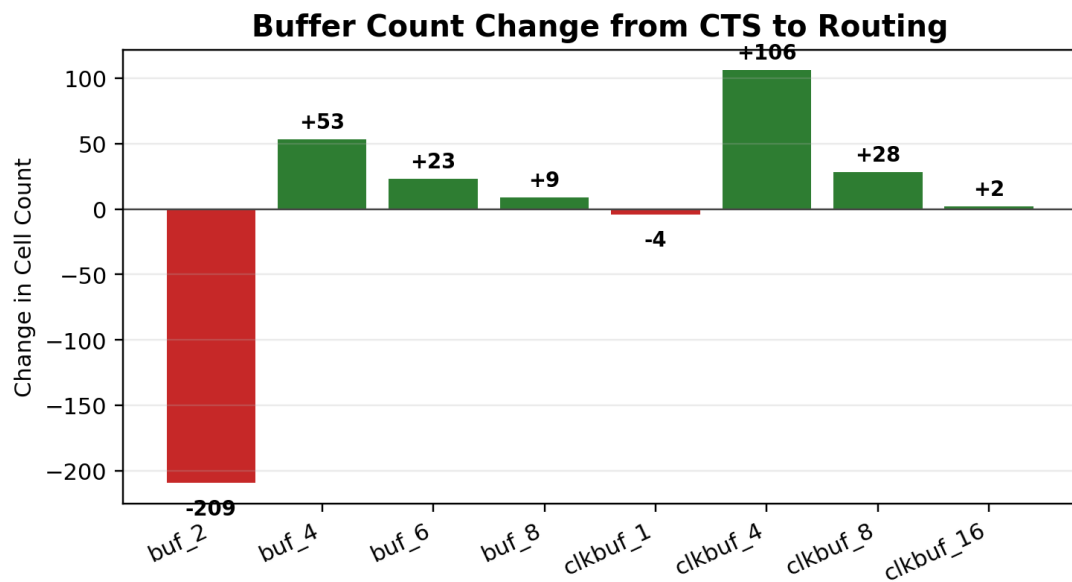


Figure: Buffer count change from CTS to routing.

Interpretation: Post-route optimization reduces many lower drive-strength buf_2 cells and increases stronger buffer variants to compensate for routed interconnect parasitics.



Inverter Comparison (CTS → Routing)

Inverter Type	After CTS	After Routing	Change
clkinv_4	1	4	+3
inv_2	79	74	-5
inv_4	1	2	+1
inv_6	0	1	+1

Cells Introduced During Routing

The following new drive-strength cells appear after routing:

Cell Type	Increase
buf_4	+53
buf_6	+23
buf_8	+9
clkbuf_4	+106
clkbuf_8	+28
clkbuf_16	+2
clkinv_4	+3
inv_4	+1
inv_6	+1

Cells Reduced/Replaced During Routing

Cell Type	Decrease
buf_2	-209
clkbuf_1	-4
inv_2	-5

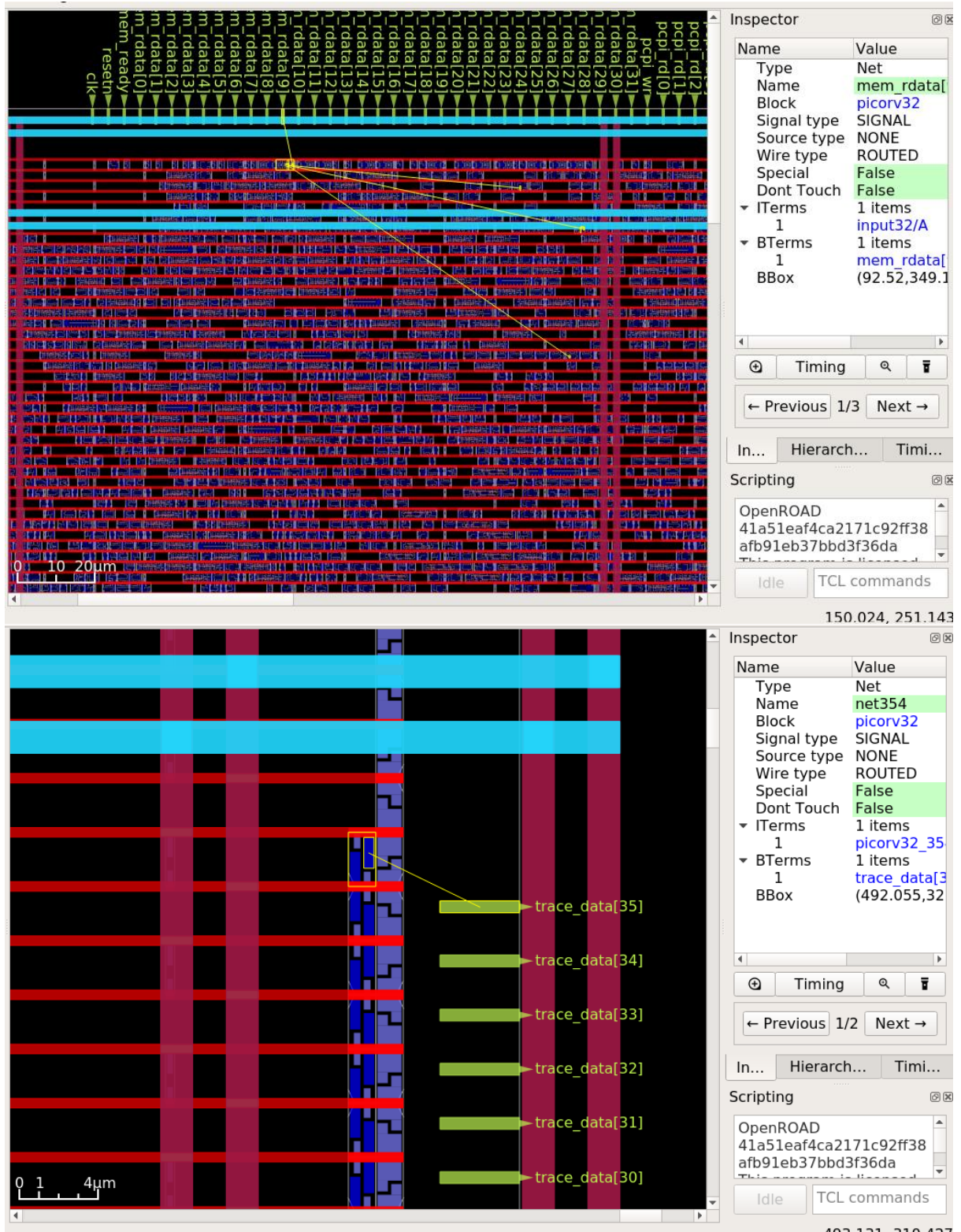
These reductions indicate that many lower drive-strength cells were resized or replaced with stronger variants during post-route timing optimization.

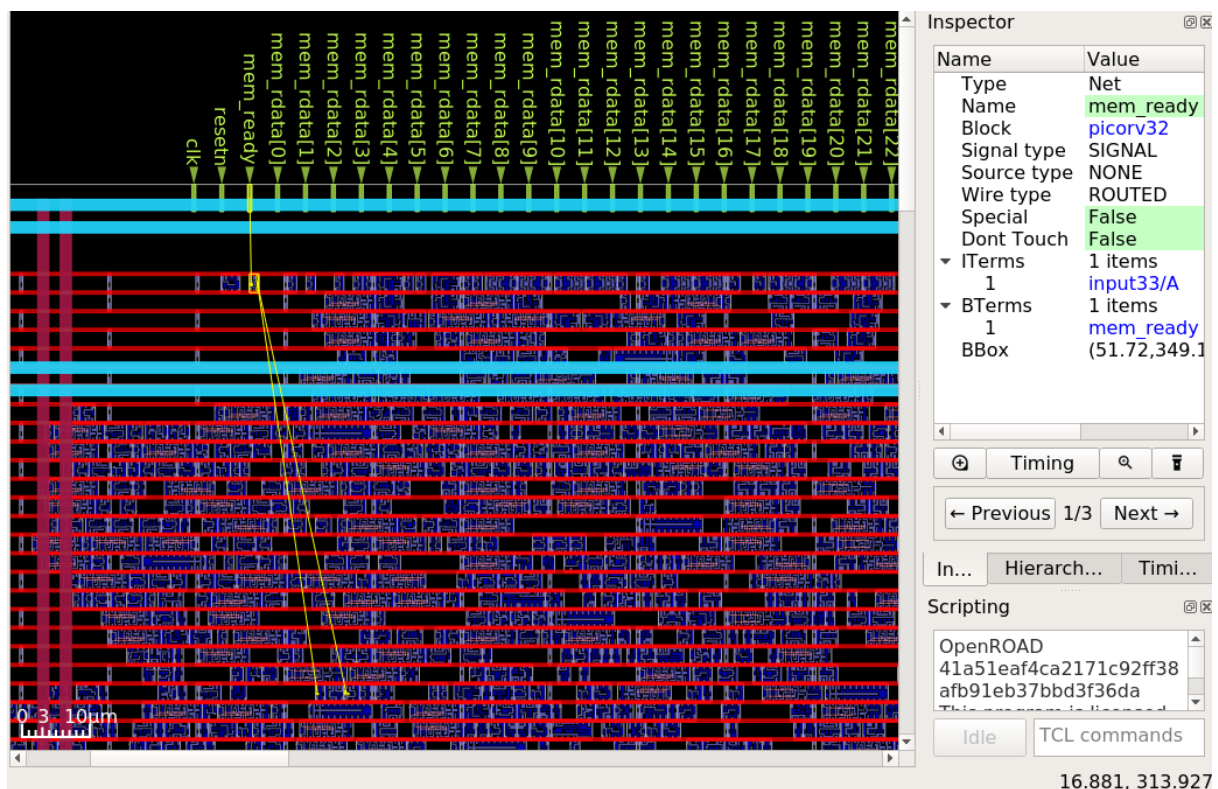
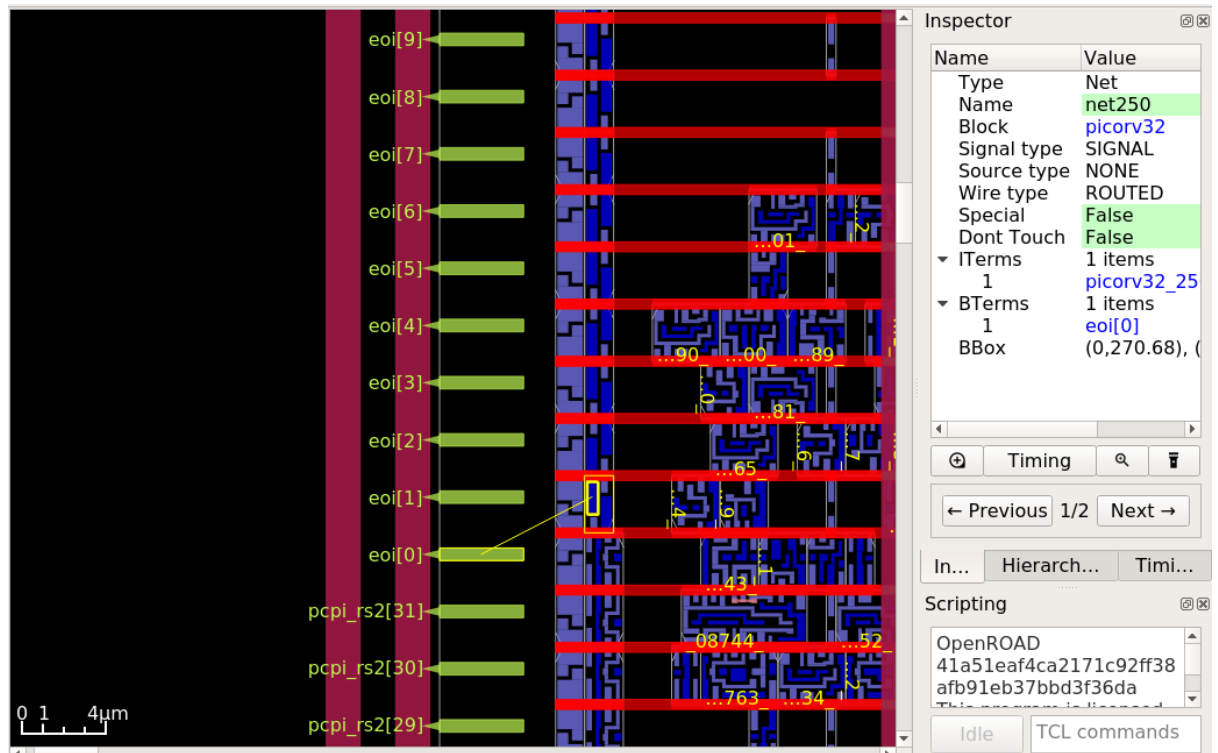
Observation for Report

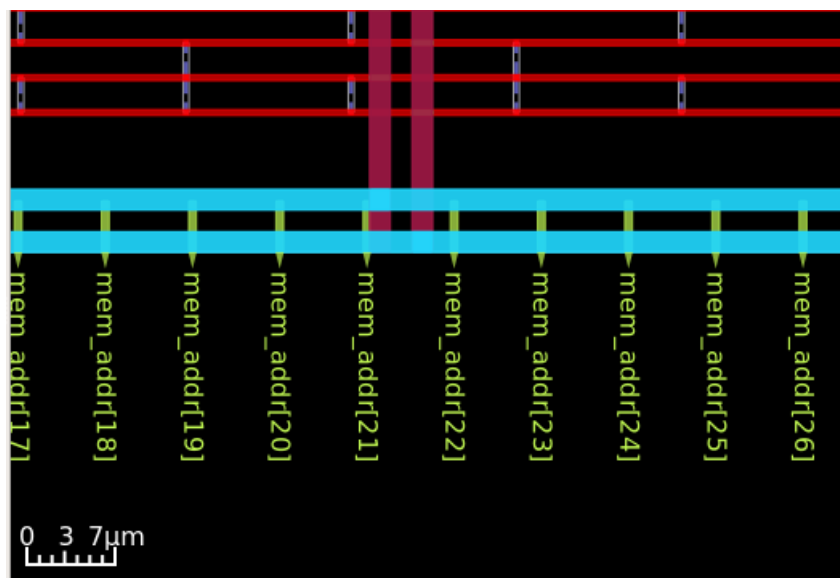
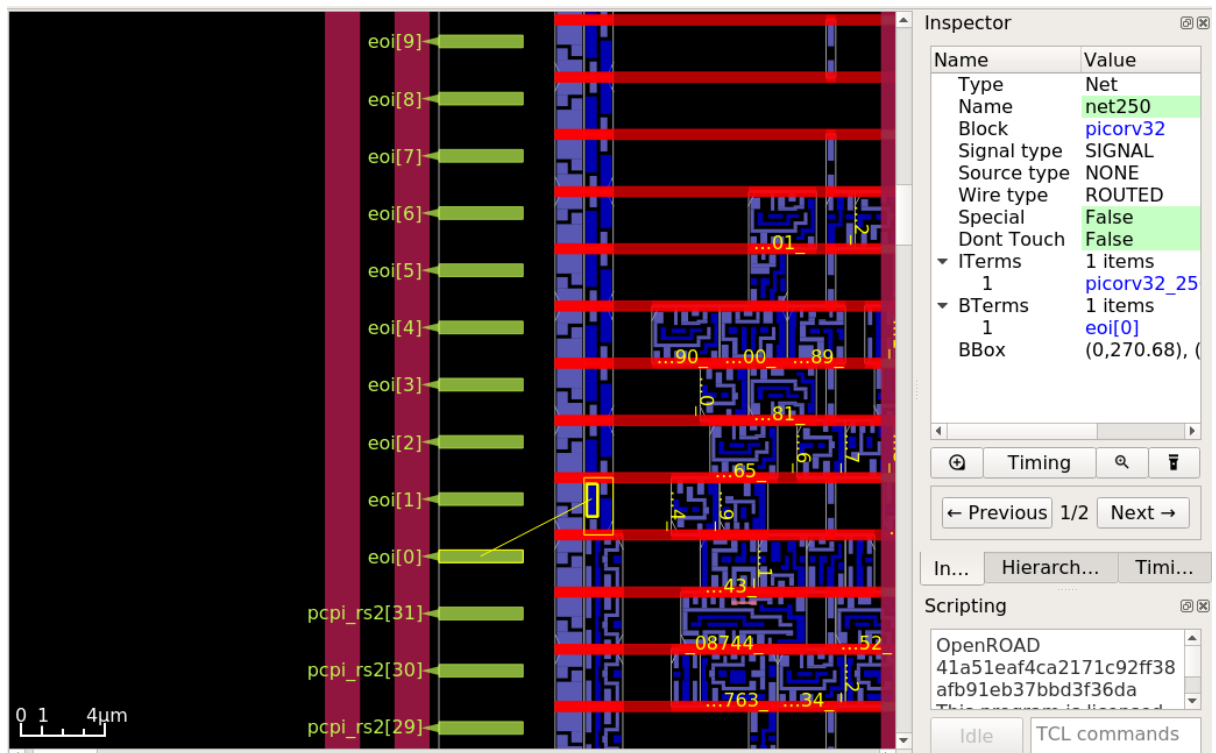


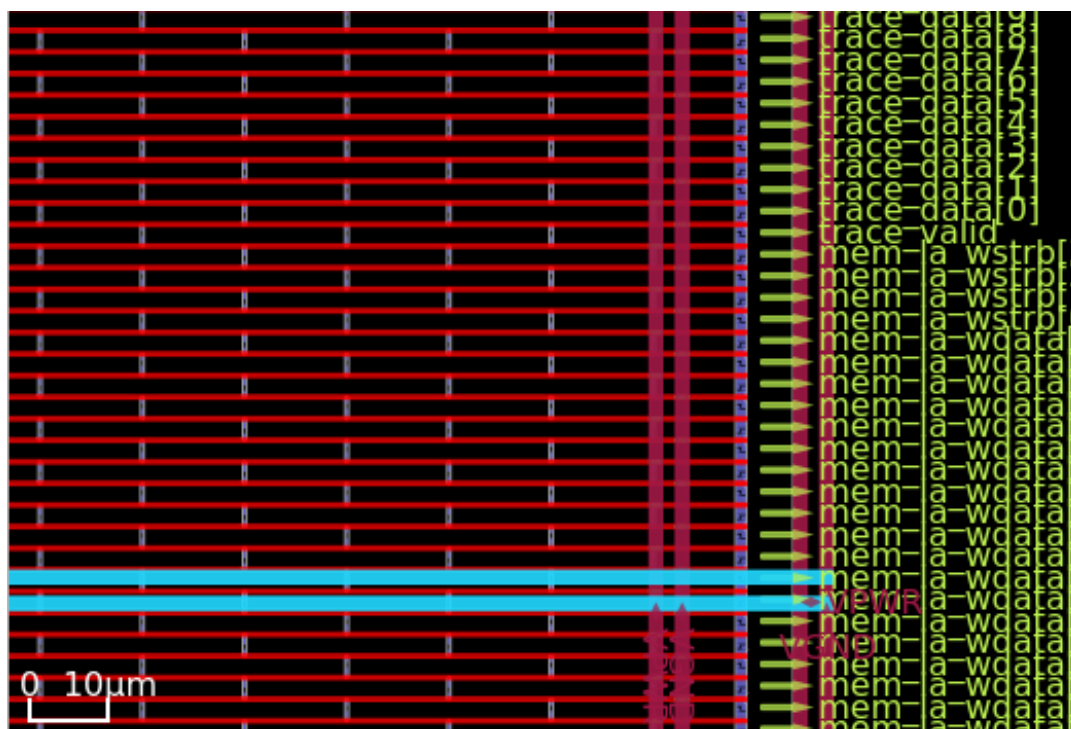
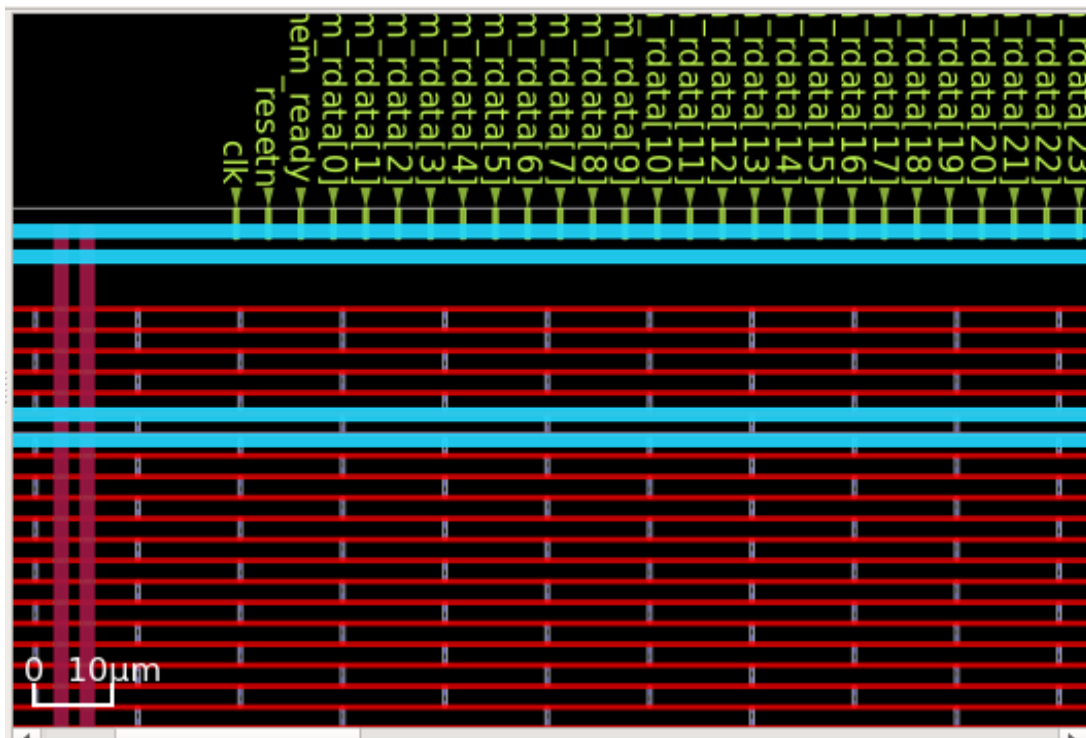
After detailed routing, the design underwent post-route optimization to compensate for the delay and parasitic effects introduced by the routed interconnects. Compared with the CTS stage, several higher drive-strength buffers such as buf_4 (+53), buf_6 (+23), buf_8 (+9), clkbuf_4 (+106), clkbuf_8 (+28), and clkbuf_16 (+2) were introduced or replaced to improve signal transition times and timing margins. Additionally, stronger inverter variants (clkinv_4, inv_4, and inv_6) appeared after routing. At the same time, the number of smaller cells such as buf_2, clkbuf_1, and inv_2 decreased, indicating that the optimization engine resized existing cells rather than simply inserting new ones. This behavior is expected during post-route optimization, where cell strengths are adjusted to meet setup, hold, slew, and capacitance constraints after considering actual routing parasitics.

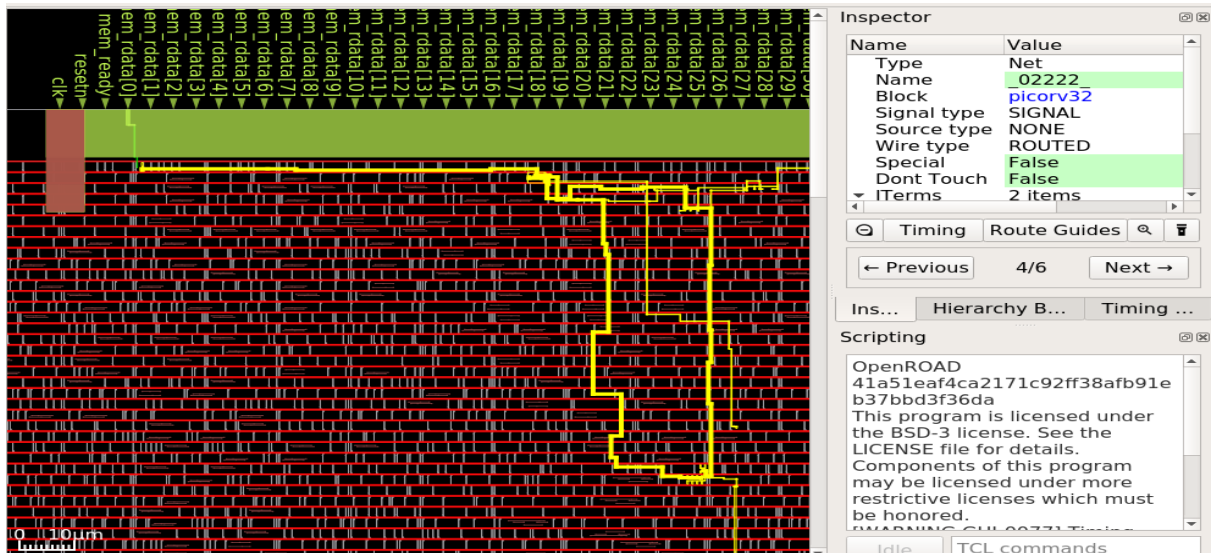
Highlight the signal routing :











Comparison of IO_PCT = 0.2 vs IO_PCT = 0.5 (Picorv32, 10 ns Clock)

Parameter	IO_PCT = 0.2	IO_PCT = 0.5	Observation
Clock Period	10 ns	10 ns	No change
Input Delay	2 ns	5 ns	Increased by 3 ns
Output Delay	2 ns	5 ns	Increased by 3 ns
Clock Uncertainty	1.5 ns	1.5 ns	No change
CTS Target Skew	0.5 ns	0.5 ns	No change
Core Utilization Target	60%	60%	No change
Aspect Ratio	0.7	0.7	No change
Core Ring	Enabled	Enabled	No change
Port Metal Layer	Metal4	Metal4	No change

Placement Timing Comparison

Parameter	IO_PCT = 0.2	IO_PCT = 0.5	Change
GPL Worst Setup Slack	+2.98 ns	-1.43 ns	Setup violation introduced
DPL Worst Setup Slack	+3.39 ns	-1.28 ns	Setup violation introduced
GPL Hold Slack	+0.17 ns	+0.18 ns	Slight improvement
DPL Hold Slack	+0.18 ns	+0.20 ns	Slight improvement
WNS	0.00 ns	-1.28 ns	Timing no longer met
TNS	0.00 ns	-2.48 ns	Multiple violating paths

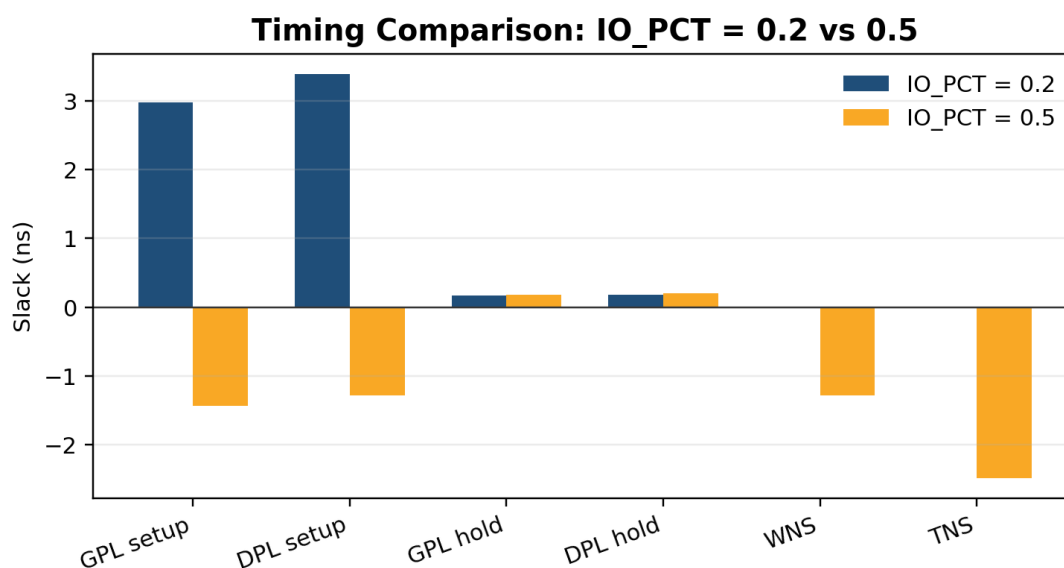


Figure: Timing slack comparison for IO_PCT = 0.2 and IO_PCT = 0.5.

Interpretation: IO_PCT = 0.5 introduces setup timing violations, while IO_PCT = 0.2 maintains clean WNS/TNS in the reported comparison.

Area and Utilization

Parameter	IO_PCT = 0.2	IO_PCT = 0.5	Observation
GPL Area	103309 μm^2	103309 μm^2	Same
DPL Area	102105 μm^2	102481 μm^2	Negligible increase
GPL Utilization	62%	62%	Same
DPL Utilization	61%	62%	Nearly unchanged

CTS Comparison

Parameter	IO_PCT = 0.2	IO_PCT = 0.5	Observation
Clock Roots	1	1	Same
Clock Sinks	1597	1597	Same
Clock Buffers Created	135	134	Difference of only one buffer
Clock Subnets	135	134	Nearly identical
Maximum Clock Tree Level	4	4	Same
Minimum Buffers/Path	2	3	Slight increase
Maximum Buffers/Path	3	3	Same



Clock Network Observations

Parameter	IO_PCT = 0.2	IO_PCT = 0.5
Clock Period	10 ns	10 ns
Clock Uncertainty	1.5 ns	1.5 ns
Clock Tree Depth	4	4
Clock Network	Same	Same
Clock Routing	Same	Same

Observation:

Changing IO_PCT did **not** significantly alter the clock tree structure. The CTS engine generated nearly the same number of clock buffers, maintained one clock root, the same number of sinks, and the same tree depth. This indicates that IO_PCT primarily affects timing constraints rather than the physical clock distribution network.

Buffer Comparison

Placement

Buffer Type	IO_PCT = 0.2	IO_PCT = 0.5
buf_1	40	39
buf_2	318	313
buf_4	231	240
buf_6	0	24
buf_8	0	7
clkbuf_1	1337	1339
clkbuf_2	0	29
clkbuf_4	281	249
clkbuf_8	74	75

CTS

Buffer Type	IO_PCT = 0.2	IO_PCT = 0.5
clkbuf_16	119	118
clkbuf_8	74	91
clkbuf_4	281	249
clkbuf_2	0	29



Observation:

Buffer distribution changes slightly due to optimization choices, but there is no major increase in total clock buffering.

Inverter Comparison

The inverter counts remain nearly identical between the two runs.

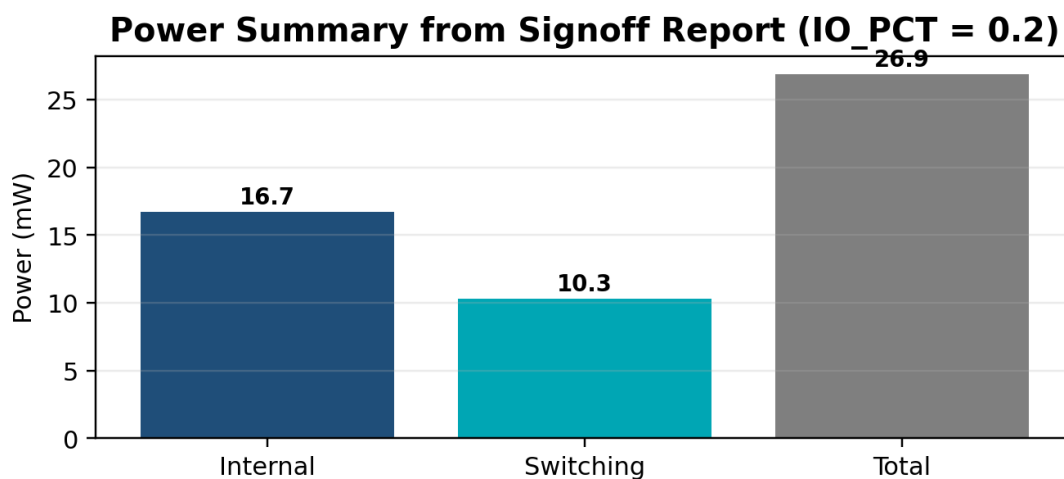
Only minor optimization differences are observed after routing.

This confirms that changing IO_PCT does not significantly affect inverter insertion.

Power Comparison

From the signoff report (shared for the IO_PCT = 0.2 run):

Power Component	Value
Internal Power	16.7 mW
Switching Power	10.3 mW
Leakage Power	45.4 nW
Total Power	26.9 mW



Leakage power: 45.4 nW (too small to show on mW scale)

Figure: Power summary from the reported signoff data.